# Algorithms and Responsibility

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**Abstract**

A long line of experimental work suggests that decision-makers can—and do—use delegation to shift responsibility for unwelcome outcomes onto an intermediary. We ask whether the same logic carries over to delegation to algorithms. In a preregistered online experiment ($N = 322$ UK Prolific participants, 161 in each role), a decision-maker makes a payoff-relevant prediction for a matched recipient with the option to delegate to an algorithm calibrated, at the individual level, to perform as well as she does. We manipulate, between subjects, whether the matched recipient can punish her for the decision. Two findings emerge. First, contrary to the preregistered prediction of the responsibility-avoidance literature, the *prospect* of punishment *reduces* rather than increases the likelihood of delegating to the algorithm (a 17 percentage-point gap, two-sided $p = 0.033$). Second, conditional on outcome, we cannot reject equal punishment of delegated and self-made decisions; the point estimate runs in the predicted direction but the test is underpowered to rule out moderate insulation effects. We discuss several candidate mechanisms for the headline reversal and argue that, among those we can evaluate, a process-ownership account—in which the moral salience introduced by the punishment possibility raises the value of being the proximate cause of a good outcome—is the most consistent with the data, while flagging that the supporting evidence is exploratory and the design cannot rule out experimenter demand. Within the bounds of the setting we study, our results offer a behavioral complement to regulatory frameworks that hold human users accountable for algorithm-mediated decisions: such frameworks may discipline algorithmic adoption rather than merely allocate post-hoc responsibility.

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# 1 Introduction

Algorithms are increasingly involved in consequential decisions: they screen job candidates, set credit scores, recommend medical treatments, evaluate parole applicants, and—more mundanely but at vast scale—route the cars driving toward this very page. As the locus of decision-making shifts toward algorithmic systems, a parallel question has moved into public, legal, and academic debate: when an algorithm-mediated decision turns out badly, who bears the blame? A natural intuition, which finds support in a long line of experimental work on responsibility avoidance through delegation, is that decision-makers can use algorithms as a means to shift blame onto a non-human intermediary. To return to a familiar example: if you and a friend arrive late because she relied on a navigation app rather than her own sense of direction, the algorithm seems a convenient scapegoat—and the next time she is unsure of the route, the prospect of avoiding your blame may itself motivate her use of the app. This paper provides experimental evidence on whether either of these intuitions is borne out.

We address two questions. *First*: can decision-makers actually shift responsibility away from themselves by delegating to an algorithm—that is, are they punished less for a delegated decision than for a self-made decision with the same outcome? *Second*: does the prospect of being held responsible motivate the decision to delegate in the first place? The two questions correspond to, respectively, the *viability* and the *motivational salience* of algorithmic responsibility avoidance. They have been studied in different settings and rarely in combination, and previous designs have struggled to identify them cleanly. We do so here in a preregistered online experiment.

Our experimental design pins down both questions. Player A first completes ten rounds of an incentivised visual prediction task (estimating a person's weight from a facial image and stated height). She then makes an final prediction whose outcome determines the payoff of a second participant, Player B: a high payoff if the prediction falls within ten pounds of the true weight, a low payoff otherwise. Player A may either make this prediction herself or delegate it to an algorithm. The algorithm is calibrated to mimic Player A's individual performance from the first ten rounds—that is, on the final prediction it produces the high outcome with the same probability as Player A herself. Player A is told this. We manipulate the punishment possibility between subjects: in the *Punishment* condition, Player B may reduce Player A's payoff conditional on the delegation decision and the realized outcome; in the *No-Punishment* condition, no punishment is possible. Player B's punishment behavior is elicited via the strategy method across all four (delegation, outcome) cells, providing a within-subject test of whether delegation insulates Player A from punishment.

Two design features distinguish our setting from the closest precedent, **?**. First, the algorithm is calibrated at the *individual* level rather than the session or group level. Existing designs typically expose subjects to a distribution of algorithmic performance and let them infer their own ability relative to that distribution; the delegation decision then carries strong signal value about the principal's self-perception, and any differential punishment can reflect overconfidence rather than responsibility shifting per se. By neutralizing relative-performance signaling at the individual level, we isolate the responsibility-shifting motive. Second, our *No-Punishment* condition turns off the punishment possibility entirely, allowing us to identify the effect of anticipated punishment on the delegation decision itself—a test that is not feasible in designs where punishment is always possible. Both features sharpen the test of responsibility shifting in algorithmic delegation that the literature has thus far been able to deliver.

Two findings emerge, and in their joint structure they push against the natural reading of the responsibility-avoidance literature. *First, the prospect of punishment reduces, rather than increases, the likelihood of delegation:* 42.5% of Player As delegate when punishment is possible, compared to 59.3% when it is not (a 17 percentage-point gap; two-sided Pearson Chi-squared $p = 0.033$, Fisher exact $p = 0.041$; the preregistered one-sided test in the H1 direction is non-significant by construction, since the result reverses the predicted direction—we discuss this in detail in Section **??**). This rejection of the standard prediction in favor of the opposite direction is the paper's main empirical contribution, and we view it as a behavioral counterpart to the regulatory intuition that holding humans accountable for algorithm-mediated decisions disciplines their use of those algorithms. *Second, conditional on outcome, we cannot reject equal punishment of delegated and self-made decisions in our setting:* Player B's punishment of Player A is statistically indistinguishable across the two, and Player A correctly anticipates this absence of differential punishment in her elicited beliefs. The point estimate of any insulation effect runs in the predicted direction but the test is underpowered against moderate alternatives ($n \approx 67$ Player Bs after attention-check exclusion); the broader claim that responsibility shifting through algorithmic delegation is altogether absent is one we cannot make with this data.

What mechanism reconciles the two findings? Among the candidates we can evaluate with our design, a *process-ownership* account is the most consistent with the data—though we treat it as a candidate hypothesis rather than an established mechanism, since the supporting evidence is exploratory and a fourth candidate (experimenter demand) cannot be ruled out by construction. A simple anticipation story—in which Player A delegates less when punishment is on the table because she expects harsher punishment for delegated decisions—is inconsistent with the elicited beliefs and with Player B's actual choices. A simple effort story—in which Player A makes the prediction herself in order to outperform the algorithm and earn the high payoff—is inconsistent with the data, in which non-delegators in the *No-Punishment* condition outperform non-delegators in the *Punishment*. What remains is the milder hypothesis that, when the moral salience of the decision is heightened by the punishment possibility, Player A wants to be the proximate cause of any good outcome for Player B, irrespective of any quantifiable belief about how she will be judged. Consistent with this interpretation, the negative correlation between Player A's task performance and her propensity to delegate is observed in the *Punishment* condition only: those most likely to deliver the high payoff want to do so personally when the moral stakes are made salient. Section **??** weighs the four candidate mechanisms in detail and is explicit about what the data can and cannot rule out.

These findings contribute to three strands of literature. First, we add to the experimental literature on responsibility avoidance through delegation (??????), by extending the canonical setting to delegation to an algorithmic intermediary in a setting with genuine outcome uncertainty. Second, we contribute to the rapidly growing literature on algorithm aversion and adoption (?????) by isolating an under-studied determinant of algorithm use: the institutional context in which the human user is held accountable. Third, and most directly, we build on ? and ? by providing a clean experimental test of two long-standing claims in the responsibility-and-algorithms literature, with a design that controls relative-performance signaling at the individual level and that includes a condition in which punishment is ruled out by construction. The contrast with ? is consequential: that study finds that decision-makers are held *less* responsible for algorithmically delegated decisions, while we find that the punishment possibility *reduces* delegation rather than increasing it. We argue in Section ?? that the two findings are reconcilable: the relative-performance signaling that drives much of Feier et al.'s differential punishment is closed off by construction in our design, isolating a different and milder margin (the motivational salience of accountability for the principal). Beyond the experimental literature, our findings speak to the regulatory debate over the assignment of responsibility for algorithmic decisions (?): within the bounds of the setting we study, holding human users accountable for algorithm-mediated outcomes appears to be a behavioral lever for direct human engagement with the decision, rather than merely a tool for ex-post moral attribution. We are explicit in Section ?? about why the present evidence—a single online experiment on a stylised prediction task with one UK Prolific sample—supports this claim only in a bounded sense, and what would be required to extend it to the high-stakes algorithmic domains (medical, judicial, hiring, financial) that motivate the regulatory debate.

The remainder of the paper is organized as follows. Section ?? reviews the relevant literature in detail. Section ?? describes the experimental design and states our two preregistered hypotheses. Section ?? presents the empirical findings, including the mechanism analysis. Section ?? discusses the implications and limitations of our results.

## 2 Related Literature

This paper bridges three strands of literature: (i) responsibility avoidance through delegation, an established line of experimental work in which the intermediary is another human; (ii) algorithm aversion and adoption, which examines when and why people are willing to use algorithmic decision aids; and (iii) responsibility attributions to decisions involving algorithms, an emerging literature at the intersection of the first two. We review each in turn, draw on a fourth body of work—identity, moral self-image, and diffusion of responsibility—to motivate our preferred mechanism, and conclude by clarifying the gap our experiment addresses.

## Responsibility avoidance through delegation

The political-science and public-choice literatures have long recognized that delegation can serve as a strategy for shifting blame onto an intermediary (**??**). Experimental economics has established the same point in incentivised settings, almost always using modified dictator games.

**?** provide the canonical experimental demonstration. In their design, a dictator may either choose between a fair and an unfair allocation of an endowment or instead delegate this choice to an intermediary whose monetary incentives are aligned with the dictator's. The recipient is adversely affected if the unfair allocation is chosen and may punish the dictator and the intermediary. Punishment is effectively shifted from the dictator to the intermediary, and the prospect of avoiding punishment constitutes a strong motive for the delegation decision itself. **?** document a complementary pattern: dictators are more generous when they allocate directly than when they select an intermediary from a set of candidates. Qualitative responsibility measures align with the punishment patterns—dictators feel less responsible when they delegate, and recipients place less blame on principals than on intermediaries.

**?** replicates the reduction in punishment under delegation but reinterprets the mechanism. He finds that punishment decreases not because of any inherent shifting of blame to the intermediary, but because the dictator no longer "directly interacts" with the affected party. This holds even when the intermediary is powerless. **?** extend the design to include various restrictions on the intermediary's choice set and confirm that punishment can be successfully avoided through delegation even to a powerless intermediary, in a setting where punishment is costly, the intermediary's incentives are aligned with the dictator's, and the intermediary can also be punished. Together, the two studies suggest that the lack of direct interaction between dictator and recipient is a key channel through which delegation reduces punishment.

**?** provides a vignette-based corroboration in a legislative-delegation context. Subjects directly evaluate blame in unincentivised hypothetical scenarios; the resulting attributions track the patterns observed in incentivised experiments, suggesting that punishment captures more qualitative blame attributions reliably. **?** push the literature in a complementary direction, showing that people are more willing to delegate decisions made on behalf of others than decisions made for themselves, especially when those decisions might have negative outcomes; the asymmetry is driven both by anticipated responsibility and by anticipated blame, and—a point we return to repeatedly—*people care more about avoiding blame for bad outcomes than about claiming credit for good ones.* **?** provides a related demonstration that people are willing to pay to delegate the act of lying, consistent with delegation reducing the psychological cost of behavior that one expects to be judged negatively. Beyond delegation per se, the broader market-design literature documents that intermediation and group decision-making can erode moral concern for third parties: **?** show that bilateral and multilateral market interactions substantially increase subjects' willingness to harm a third party (a mouse) for monetary gain, a canonical demonstration of diffusion of responsibility.

These studies share a common conclusion: in settings with self-interested choice between a fair and an unfair allocation, delegation reduces punishment of the principal, and the prospect of punishment in turn motivates delegation. Their design space, however, has two important features that constrain external validity to algorithmic settings. First, the intermediary is human, and the principal can typically be punished for choosing a particular kind of intermediary. Second, the underlying choice is between a known fair and a known unfair allocation, with no inherent uncertainty in the outcome. Whether responsibility avoidance is just as effective when the intermediary is an algorithm and the underlying decision involves genuine uncertainty is the question we take up. We turn next to what is known about delegation when the intermediary is algorithmic.

## Delegation to algorithms

A large literature studies when people use algorithmic decision aids. **?** introduced the term "algorithm aversion" for the finding that people prefer their own (or other humans') judgement over algorithmic predictions, even after observing that algorithms outperform human decision-makers. Subsequent work has documented many moderators of this preference. Algorithm aversion is stronger when the algorithm cannot be observed to learn from its mistakes (**?**), when the algorithm's reasoning is opaque (**??**), when the algorithm is presented as more complex (**?**), and—most importantly for our purposes—when the underlying decision has moral content or affects third parties (**???**). Algorithm aversion is also stronger when the task is perceived as more uncertain (**?**), when delegation removes any ability to override the algorithm's choice (**?**), and when the algorithm is used in tasks perceived as subjective (**??**). **?**, by contrast, document conditions under which subjects show *appreciation* of algorithmic advice over human advice; the picture that emerges is therefore one of substantial heterogeneity in algorithm preferences across tasks, contexts, and elicitation methods. Recent additions include **?**, who show that loss-framing of decision tasks eliminates the preference for human assistance over AI assistance, and **?**, who develop a menu-based design to elicit willingness to cede control to algorithms versus humans. **?**, **?**, and **?** provide systematic reviews of this literature.

Within this body of work, a smaller subset focuses on settings in which the decision being delegated has consequences for someone other than the decision-maker, sometimes called the *moral domain* (**?**). **?** find that subjects in a laboratory experiment preferred to delegate a simple calculation task to another participant rather than to an algorithm even when the task was a straightforward arithmetic exercise where algorithmic advantage should have been salient; subjects had observed the algorithm fail in 80% of cases, however, which complicates inference about pure preferences. **?** report similar aversion when the decision affects both the decision-maker and a third party, and **?** document the same in a financial decision-making context. **?** extend the picture to the human-resources domain, showing that algorithmic HR decisions are perceived as less fair than identical human decisions because of "algorithmic reductionism"—the perception that algorithms quantify and decontextualise the qualitative features of performance.

The consensus across this subset is that aversion to algorithms is amplified when third-party welfare is at stake. The mechanism, however, is unsettled: *lack of trust in the algorithm*, *loss of perceived control*, and *anticipated reputational or moral costs* have all been proposed, and existing designs typically cannot separate them. We argue in Section **??** that one specific mechanism—*anticipation of judgement when punishment is possible*—deserves a sharper look, since our results show that turning the punishment possibility on or off substantially shifts adoption.

## Responsibility perceptions of decisions involving algorithms

A small but growing literature examines how decisions involving algorithms are evaluated by third parties. Most early work is hypothetical and unincentivised: subjects read scenarios involving human-algorithm pairs and rate responsibility on Likert scales. **?** find that observers attribute less responsibility to a company when an accident involves technology. **?** find similarly that observers attribute less fault to a doctor who follows a (bad) recommendation from an automated decision aid. **?** replicate this effect in a contemporary setting. **?** consider moral violations produced by human-algorithm pairs and find that humans who monitor algorithms are blamed less than humans acting alone or humans receiving algorithm recommendations, suggesting that responsibility attributions are sensitive to the precise division of authority between human and machine. **?** document a robust "algorithmic outrage asymmetry": moral violations by algorithms produce less outrage than identical violations by humans. **?** extend the discussion to liability law and find experimentally that physicians who follow standard-care AI recommendations face reduced legal exposure; they argue that this could constitute an incentive to follow such recommendations specifically because doing so shields physicians from blame. The literature thus far suggests that responsibility attributions to humans involved with algorithms differ systematically from attributions to humans acting alone, but the direction depends on the exact configuration.

A smaller incentivised experimental literature has begun to test these ideas in laboratory settings with monetary stakes. **?** compare a modified dictator game in which the dictator either acts alone or shares responsibility with a computer; they find no significant differences in self-reported responsibility between human-computer and human-human teams, although a (statistically insignificant) tendency towards more selfish allocations under shared responsibility with humans. **?** find in a computer-aided HR setting that algorithm aversion *dominates* blame avoidance: the prospect of using an algorithm to delegate an unpleasant HR decision (a dismissal) does not overcome the underlying aversion to algorithms in this domain. The most closely related study is **?**, who study delegation to an algorithm in an incentivised laboratory experiment. In their design, subjects first complete ten logic tasks, are randomly assigned to either a "human" or "machine" treatment, and then either rely on their own performance or delegate to (respectively) another human or an algorithm to determine a third party's payoff. Third parties may then reward or punish. The headline finding is that decision-makers are held *less* responsible for decisions delegated to an algorithm than for decisions delegated to another human, conditional on outcome. Two recent additions push this literature further. **?** extend the blame chain to include the programr behind the algorithm: in their design, AI programrs and the delegator can both escape responsibility for inegalitarian AI-mediated allocations, creating a "moral wiggle room" that incentivises programrs to push the AI toward more inegalitarian choices. **?**, in a high-profile recent study in *Nature*, document that humans request unethical behavior from machine agents more readily than from human agents, in part because the machines (large language models including GPT-4, Claude 3.5 Sonnet, and Llama 3.3) comply with such instructions 58–98% of the time, compared to 25–40% for human agents. Their setting (active dishonesty under instruction) differs from ours (delegation under outcome uncertainty), but the AI-as-blame-shield motif is shared.

The **?** design, which is the immediate precedent for ours, has three features that complicate the inference about pure responsibility shifting. First, the algorithm in their experiment is calibrated at the session level rather than at the individual level: each subject is shown the distribution of algorithmic performance across the relevant pool, and her delegation decision then reflects beliefs about her own ability *relative* to that distribution. The delegation choice therefore carries strong signal value about Player A's self-perception, which contaminates downstream punishment as a measure of responsibility shifting per se. **?** indeed report that the delegation decision is well explained by perceived own performance relative to the algorithm. Second, the same subject acts as both delegator and evaluator, introducing additional motives—self-justification, hypocrisy aversion, projection—into the punishment decision. Third, their design does not include a treatment in which punishment is ruled out by construction, so the question of whether the punishment possibility itself motivates delegation cannot be addressed in their data.

## Theoretical microfoundations: identity, moral self-image, and process ownership

The mechanism we adopt in interpreting our results—a desire to be the proximate cause of a good outcome that is amplified by the moral salience of accountability—draws on two adjacent theoretical traditions. The *identity / moral self-image* literature in economics, anchored by **?**, models people as caring about "who they are" and inferring their own values from past choices. Identity functions as an asset that responds non-monotonically to acts and threats, producing asymmetric reactions to good versus bad outcomes that map naturally onto the credit-blame asymmetry documented by **?**. The *psychological-ownership* literature, in turn, has documented that customising or actively choosing an output transfers a sense of ownership to it: people feel more responsible for outcomes they directly bring about than for those they delegate. The intersection of these two literatures with our setting is largely unexplored: no formal model directly links the identity framework to algorithm-mediated decisions. Our M3 mechanism (developed in Section **??**) is in this sense a hypothesis derived from the identity framework rather than a confirmed account, and we treat it as such throughout.

## Where this paper fits

Our experiment is the first to our knowledge to (i) calibrate the algorithm to mimic each individual subject's performance, eliminating the relative-performance confound; (ii) separate the role of delegator and evaluator across distinct subject pools; and (iii) include a between-subject manipulation that turns the punishment possibility on or off, providing a clean test of whether anticipated punishment is a motive in the delegation decision itself. The first two features sharpen the test of the responsibility-shifting hypothesis (H2) by removing alternative interpretations of differential punishment. The third moves beyond the existing literature, which has documented punishment differentials *conditional on* the punishment possibility but has not directly identified whether the threat of punishment causally moves the delegation decision.

**Reconciling our findings with ?.** Two of our results sit in apparent tension with the closest precedent. First, **?** document that decision-makers are punished *less* for algorithmically delegated decisions; we find no significant punishment differential. Second, the natural prediction from **?**'s own findings is that the prospect of punishment should make algorithmic delegation *more* attractive; we find the reverse. We do not view these contrasts as contradictions, but as evidence that the design features which differ between the two studies are doing real work. The differential punishment in **?** is most plausibly read as third parties extracting a signal from the delegation decision about the principal's self-perceived ability relative to the algorithm—a signal that is maximally informative when (as in their design) the algorithm is calibrated at the session level and when the principal's relative ability is uncertain. Our individual-level calibration, made common knowledge, closes off this signal-extraction channel by construction: a Player B in our setting has no basis to read delegation as overconfidence, laziness, or ability avoidance. The disappearance of the punishment differential in our data is therefore consistent with, rather than refutes, the signal-extraction reading of **?**'s result. By the same logic, the responsibility-shifting motive that **?**'s setup leaves on the table is muted in ours, isolating a different and milder margin: the moral salience of accountability for the principal herself, independent of any third-party inference. The reversal we observe in the delegation decision speaks to that margin—and we are explicit that the question of which design feature is causally responsible for the sign reversal cannot be resolved without an experiment that varies relative-performance calibration directly, which we do not do here.

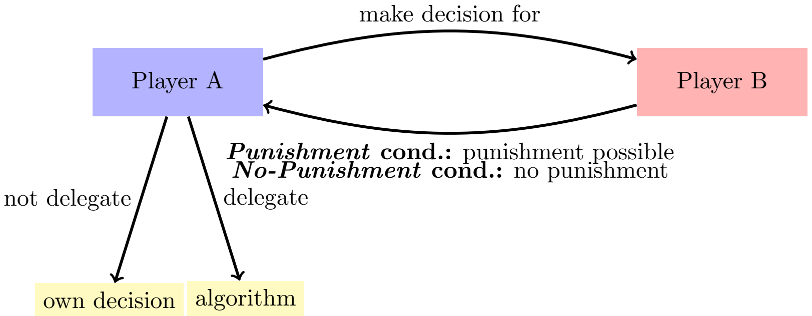
We see scope, then, for the two studies to be parts of a single picture: **?** identifies the signal-extraction channel through which delegation can shift third-party blame; we identify a process-ownership channel through which the prospect of accountability can pull the principal toward direct engagement with the decision. The two channels are not mutually exclusive. **?**, looking at a third margin (the programr behind the algorithm), and **?**, looking at a fourth (the compliance asymmetry between human and machine agents on instructed unethical tasks), populate further nodes in this expanding map. A unified design that varies the relative-performance calibration, the punishment possibility, and the visibility of the algorithm's compliance properties would be the natural next step. We discuss the implications for future work in Section **??**.

# 3 Experimental Design

We conducted an online experiment, preregistered at AsPredicted #133980 (aspredicted.org/hs9az8.pdf), on the British platform Prolific between February and June 2023, recruiting 322 participants from the United Kingdom. The design combines a between-subject manipulation of whether costly punishment is possible with a within-subject elicitation of conditional punishment behavior.

Within each condition, participants assumed one of two roles. Player A decides whether to make a payoff-relevant prediction for Player B directly, or to delegate the prediction to an algorithm calibrated to perform as well as she does. Player B is the matched recipient whose payoff is fully determined by the outcome of Player A's prediction; in the *Punishment* condition Player B may subsequently punish Player A, conditional on both the delegation decision and the realized outcome, while in the *No-Punishment* condition no punishment is possible. Figure **??** summarizes the basic structure.

Figure 1: Conditions overview




We targeted a balanced design of 80 participants per role per condition, for a planned sample of 320 subjects.[1] The experiment was programmed in oTree (**?**) and run via Prolific. Eligibility followed Prolific's standard quality screens—minimum 95% prior approval rate, no warnings on file, and a fluent-English filter—together with a desktop-only restriction (no mobile or tablet) to keep the screen layout consistent across subjects.

[1]Appendix **??** reports the full ex-ante minimum-detectable-effect calculations underlying this cell size.

## Procedure

### Player A

After providing informed consent, Player A completed ten incentivized rounds of a visual prediction task. In each round, she predicted a person's weight from a facial image and the person's height. Predictions could be entered via slider or typed directly in kilograms, pounds, or stones-and-pounds; no feedback was provided between rounds.[2] At the end of the experiment, one of the ten rounds was randomly selected for payment: Player A earned £4 if her prediction in that round was within 10 lbs of the true weight and £2 otherwise. These initial rounds served three purposes simultaneously: they established a stake from which any later punishment could be deducted; they familiarized Player A with the task; and they generated the performance data needed to calibrate the algorithm.

The choice of a visual prediction task is deliberate. Algorithm preferences depend sensitively on the algorithm's perceived relative competence (**??**);

FB CHECK: Verify that Dietvorst (2015) and Logg (2019) support the "perceived relative competence" framing. Dietvorst (2015) is largely about confidence loss after observing errors rather than about ex-ante perceived competence; Logg (2019) is about algorithm appreciation as a function of perceived ability.

as described below, our design neutralizes this concern by calibrating the algorithm to match each subject's individual performance. The visual nature of the task makes algorithmic error plausible and lends credibility to the equal-performance claim, in contrast to purely mathematical tasks where subjects may be skeptical that an algorithm could ever fail (**?**).

FB CHECK: Verify that Gogoll & Uhl (2018) actually support the math-task scepticism contrast. The paper's main claim is about moral-domain aversion; the math-task framing may not be in the paper.

Predicting body weight from a facial image has been used in earlier experiments (**?**) and arguably levels the playing field between human and algorithm: humans have a lifetime of practice at this mapping, while algorithms cannot easily exploit large training corpora here. Finally, we chose a prediction task involving genuine uncertainty rather than a setting with an obvious self-serving motive (as in classic dictator-game variations). This reflects the nature of many real-world applications of algorithms—financial forecasting, medical diagnosis, fraud detection—in which outcomes are inherently uncertain and performance is evaluated ex post.

After the ten rounds, Player A was randomly paired with another participant, Player B. Player A was informed that her final prediction would not affect her own payoff but would almost entirely determine the payoff of Player B, who had not completed the prediction task

[2] All individuals depicted in the images had explicitly consented to the use of their photographs for research purposes.

himself: if the prediction fell within 10 lbs of the true weight, Player B would receive a high payoff (£5); otherwise a low payoff (£1). Player A then learned that she could either make the prediction herself or delegate it to an algorithm calibrated to mimic her own performance from the first ten rounds, so that on expectation the algorithm would produce the high payoff for Player B at the same rate as Player A would.

Concretely, the algorithm draws a Bernoulli outcome with success probability equal to Player A's empirical accuracy rate over her first ten rounds. A subject who scored 3/10 in the initial rounds thus delegates to an algorithm that produces the high payoff for Player B with 30% probability. This individual-level Bernoulli calibration is the central design innovation relative to the closest precedent (**?**), and it does substantive work in the identification argument. First, it ensures the algorithm's success probability equals Player A's empirical accuracy over the first ten rounds for every subject, making the equal-performance claim true in the data on which the calibration is based. Second, it strips any signal value that the delegation decision might otherwise carry about Player A's beliefs concerning her own ability relative to the algorithm: a Player B observing the delegation decision has no basis on which to read it as overconfident, lazy, or selfish. This closes off the most plausible third-party signal-extraction channel through which delegation might otherwise insulate the principal from punishment.

The contrast with session-level calibration is consequential. Had the algorithm instead been calibrated to reflect aggregate performance across a pool of participants—as in **?**—the delegation choice would be confounded by each subject's beliefs about her own ability relative to the pool. Indeed, **?** report that the delegation decision in their setting is largely driven by perceived own performance relative to the algorithm. Pairing the algorithm to each subject individually, and making this common knowledge between Player A and Player B, strips this confound out by construction.

In the *Punishment* condition, Player A was further informed that Player B could reduce her payoff by up to £2, with the deduction depending on both her delegation decision and the realized outcome. In the *No-Punishment* condition, Player B had no punishment option. The two delegation options (own decision / algorithm) were presented in random order across subjects. We attached no monetary cost to delegation. This brings us closer to Player A's intrinsic preference over delegation in the *No-Punishment* condition, since any positive cost would conflate the responsibility-shifting motive with a price effect; costless delegation is standard in the responsibility-shifting literature for this reason (**???**).

We then asked Player A to report her beliefs about Player B's punishment in each of the four scenarios resulting from the combination of delegation decision (yes/no) and prediction outcome (high/low payoff for Player B). Beliefs were elicited on the same scale as actual punishment (£0–£2 in £0.10 increments) and the four scenarios were presented in block-wise random order. We chose not to incentivize belief elicitation for three reasons. First, only two of the four scenarios remain payoff-relevant after the delegation decision, making the others purely hypothetical. Second, an incentivized mechanism could induce hedging behavior.

Third, an incentive-compatible scoring rule across four scenarios would have substantially increased task complexity. Because we are nonetheless interested in comparing punishment beliefs to actual punishment, we included an attention check on the belief-elicitation screen.[3] In the *No-Punishment* condition, no punishment was possible; we elicited beliefs about hypothetical punishment to maintain consistency across treatments.

Player A was then asked to assess her own performance in the first ten rounds by assigning a probability mass (summing to 100%) to each possible score from 0 to 10. We elicited this belief after the delegation decision, and after Player A had learned of the equal-performance calibration, for three reasons: the post-update belief is the analytically relevant input for any subsequent behavior; pre-decision elicitation would risk introducing measurement-induced anchoring on the delegation decision itself; and the elicitation is used as a moderator rather than as a determinant of delegation in the primary analysis. The elicitation was incentivized: Player A earned an additional £0.30 with the probability she had assigned to her actual score.[4] Player A was also asked to estimate the population distribution of scores by assigning a share to each possible score from 0 to 10, which lets us construct a measure of perceived relative ability that the prior literature's pooled-calibration designs cannot identify. One score was randomly drawn at the end of the experiment and Player A earned £0.30 if her estimate for that score was within five percentage points of the realized population share.

Finally, we elicited risk preferences using an incentivized multiple price list in the style of **?** and administered a short questionnaire on socio-demographic characteristics and technology affinity, before presenting Player A with her preliminary payoff. Figure **??** in Appendix **??** summarizes the experimental timeline for Player A.

### Player B

After providing informed consent, Player B was told that he had been randomly matched with a Player A and was introduced to the task Player A had completed. He learned that his own payoff would be determined by the outcome of Player A's final prediction—£5 if it fell within 10 lbs of the true weight and £1 otherwise—while Player A's payoff depended solely on her performance in the initial ten rounds. Crucially, Player B was also told that Player A could delegate the final prediction to an algorithm calibrated to perform equally well as Player A in the earlier rounds, and that Player A herself was aware of this equal-performance calibration. The equal-performance claim is therefore common knowledge between the two players, a feature on which our identification argument rests.

In the *Punishment* condition, Player B could choose to reduce Player A's payoff by up to £2 in £0.10 increments. We focus on punishment rather than reward because the responsibility-attribution literature has predominantly operationalized blame through sanction-imposing be-

[3]Subjects who failed the attention check are excluded from belief-related analyses.

[4]The mechanism is incentive-compatible only for risk-neutral subjects; in practice most subjects spread probability mass across several scores.

havior (???), and because the applied policy questions about algorithmic accountability that motivate our paper are likewise framed in sanction-language. Imposing strictly positive punishment in any cell required Player B to forgo £0.10 of his own earnings for that cell, ensuring that punishment was costly to the punisher.[5] We elicited punishment decisions for all four combinations of delegation (yes/no) and outcome (high/low payoff for Player B) via the strategy method, implementing only the decision matching the realized combination; the strategy method gives us the within-subject power to detect cell-level differences with the achieved sample size.[6] Because the punishment decision is the central outcome variable, we included an attention check on the punishment-elicitation screen.[7] In the *No-Punishment* condition, where actual punishment was not possible, we elicited hypothetical punishment decisions on the same scale to keep the experimental flow comparable across conditions. Player B's perceptions of task difficulty, his risk preferences, and a short questionnaire followed, before he was presented with his final payoff. Figure ?? in Appendix ?? summarizes the experimental timeline for Player B.

Once Player B's decisions had been recorded, Player A's final payoff was calculated and all participants were paid on the day of participation. Figure ?? in Appendix ?? reports the experimental flow of participants by condition and role, including recruitment, attention-check exclusion, and the analysis sample.

## Hypotheses and identification

Our design tests two preregistered hypotheses, both grounded in the literature on responsibility avoidance through delegation (??????) and on responsibility attributions to algorithmic decision-makers (???).

**H1 (Delegation):** The introduction of potential punishment increases the likelihood that Player A delegates the decision to the algorithm.

H1 is grounded in empirical evidence that responsibility avoidance shapes delegation decisions involving human intermediaries. ? also show that subjects are more likely to delegate morally consequential decisions even to a die roll—an early form of delegation to a non-intentional agent—suggesting that responsibility avoidance persists when the agent lacks intentionality. Whether the same logic extends to *algorithmic* delegates is less settled. ? report that delegation rates are similar for human and algorithmic intermediaries, suggesting that the *kind* of intermediary may matter less than the mere availability of one—but their design does not

[5]The cost is fixed per scenario rather than proportional to the punishment level, so that strictly positive punishment is always costly relative to no punishment but the cost does not increase with the severity of the sanction.

[6]The strategy method may yield different absolute punishment levels than direct elicitation; we discuss the implications and a possible direct-method companion as future work in Section ??.

[7]The check asked the subject to confirm a numerical aspect of the payoff structure that had been displayed on the previous screen. Subjects failing this check are excluded from punishment-related analyses.

directly compare delegation rates across punishment regimes, leaving open the question H1 addresses.

FB CHECK: Felix asked whether Feier et al. (2022) report whether delegation propensity is HIGHER under punishment than without for both AI and human delegates. Their design does not appear to directly run this between-treatment comparison, but please verify in the paper.

H1 is identified by the between-subject comparison of Player A's delegation rate across the *Punishment* and *No-Punishment* conditions, which exploits the punishment manipulation as the only between-subject difference and holds fixed every other feature of the decision environment.

**H2 (Punishment):** Conditional on a low-payoff outcome for Player B, Player A is punished less when she delegated the decision to the algorithm than when she made it herself.

H2 mirrors H1 on the punishment side. **?** find that delegation to a randomization device still allows decision-makers to avoid some punishment, although less so than delegation to another human. **?** find that decision-makers are rewarded less when a negative outcome is directly attributable to their own action, compared to when it results from an algorithm to which they delegated. Their study differs from ours in three respects, however: a different task; an emphasis on reward differentials rather than punishment; and—most consequentially—a design that does not control for relative performance at the individual level. As a result, in their setting the delegation decision carries a strong signal about Player A's beliefs concerning her own ability relative to the algorithm; a failure to delegate may then be punished as overconfidence rather than as direct responsibility-taking, contaminating the inference about responsibility-shifting per se. H2 is identified by the within-subject comparison across the four (delegation, outcome) cells of Player B's strategy-method punishment, which isolates the conditional-on-outcome effect of delegation on punishment, holding fixed both the realized consequences for Player B and all between-subject heterogeneity in Player B's punishment proclivity.

Two further design features deserve note. First, although the prediction task affects Player B's payoff, the incentives of the two players are nominally aligned: Player A has no direct financial interest in Player B receiving the low payoff, and a successful prediction strictly weakens her own expected punishment.[8] This alignment mirrors canonical real-world settings for algorithmic delegation—medical diagnosis, financial forecasting, judicial decision-support—in which the decision-maker has no direct material interest in the affected party's outcome but bears reputational or moral consequences for poor outcomes; by replicating this alignment, our design speaks directly to the policy-relevant cases. A standard intention-based punishment motive should therefore have less bite here than in dictator-game settings where the decision-maker can choose to be selfish. Second, although the underlying decision (a prediction under uncertainty) is not itself moral, the setup nonetheless qualifies as part of the *moral domain* as defined

[8]In the *Punishment* condition, a better outcome for Player B reduces expected punishment for Player A; in the *No-Punishment* condition there is no punishment to reduce. Either way, Player A's incentives are weakly aligned with Player B's.

in the algorithm-aversion literature (?): Player A's choice imposes monetary externalities on a matched recipient, and a poor decision causes real, albeit modest, monetary harm. The implications of these two design choices for the interpretation of our results are taken up in Section ??.

# 4 Results

**[General - Sample Description: Timing of experiment, selection criteria, time to completion, earnings, balance across observables]** Data collection took place between February and June 2023 through the British online platform Prolific, with all participants recruited from the United Kingdom.[9] The experiment was programmed in oTree [reference]. A total of 322 subjects were recruited (161 assigned to Player A, 161 to Player B). [10] On average, Player A spent 12 minutes completing the experiment and earned YZ, while Player B spent approximately 8 minutes and earned YZ. Treatment randomization was effective, as demonstrated by the balance across observable characteristics (Appendix ??, Table ??). [Balance is only important for Player A, because there we do comparisons across treatments, for Player B it is about a within-subject decision. Balance across observables is only given for Player A.] [for external validity concerns: only people at pc, no phone or tablet]

Data collection took place between February and June 2023 through the British online platform Prolific, with all participants recruited from the United Kingdom.[11] A total of 322 subjects were recruited (161 assigned to Player A, 161 to Player B).[12] On average, Player A spent 12 minutes completing the experiment and earned an average bonus of £2.49 on top of the Prolific show-up fee, while Player B spent approximately 8 minutes and earned an average bonus of £2.30. Treatment randomization was effective, as demonstrated by the balance across observable characteristics (Appendix ??, Table ??). A flow diagram showing recruitment, attrition, attention-check exclusion, and the analysis sample, broken down by condition and role, appears in Appendix ??; in particular, attrition rates do not differ meaningfully across the *Punishment* and *No-Punishment* conditions.

We preregistered exclusion of subjects who failed the attention check on the screen at which their main outcome variable was elicited. For Player A, this is the belief-elicitation screen; for Player B, the punishment-elicitation screen. Of 161 Player Bs, 135 pass the attention check (83.9%); the corresponding pass rate for Player A is 131/161 (81.4%). All main results are robust to including subjects who fail the attention check; we report these robustness checks in Appendix ??.

[9] EXCLUSION CRITERIA

[10] We preregistered the collection of data from 320 participants (80 subjects per role per treatment). Due to attrition and subsequent re-matching we recruited an extra pair in the treatment condition.

[11] EXCLUSION CRITERIA: leave a comment for Felix to insert later

[12] We preregistered the collection of data from 320 participants (80 subjects per role per treatment). Due to attrition and subsequent re-matching we recruited an extra pair in the *Punishment* condition.

FB: maybe move to first set of results. Also the appendix does not show anything and the default should be including everyone.

## Player A – Delegation behavior

**[Introduction of Main result 1 - Punishment prospect reduces likelihood to delegate - Non-parametric tests - Figure]** We begin by examining the delegation behavior of Player A. In the baseline condition, where punishment by Player B was possible, 42.5% of decisions were delegated to the algorithm. In contrast, in the treatment condition, where Player A cannot be punished by Player B, 59.3% of decisions were delegated to the algorithm (Figure **??**). Thus, the introduction of the punishment possibility significantly reduces the likelihood to delegate the decision to the algorithm (Chi-squared test p-value: 0.049, Fisher-exact test p-value: 0.041). [Those are the p-values for the two-sided test, but we may have a one-sided hypotheis. Should we instead have equal punishment and then reject that? Do you make the hypothesis rejectable? Moreover, STATA gives other p-values]

**Headline.** In the *Punishment* condition, where Player B can punish, 42.5% of Player As delegated the decision to the algorithm. In the *No-Punishment* condition, where punishment is impossible, 59.3% delegated. Figure **??** illustrates the difference. The introduction of potential punishment thus significantly *reduces* the likelihood of delegation to the algorithm (Pearson Chi-squared, $p = 0.033$; Fisher's exact, $p = 0.041$).[13][14]

**Result 1:** The possibility of punishment significantly reduces the likelihood that Player A delegates the decision to the algorithm.

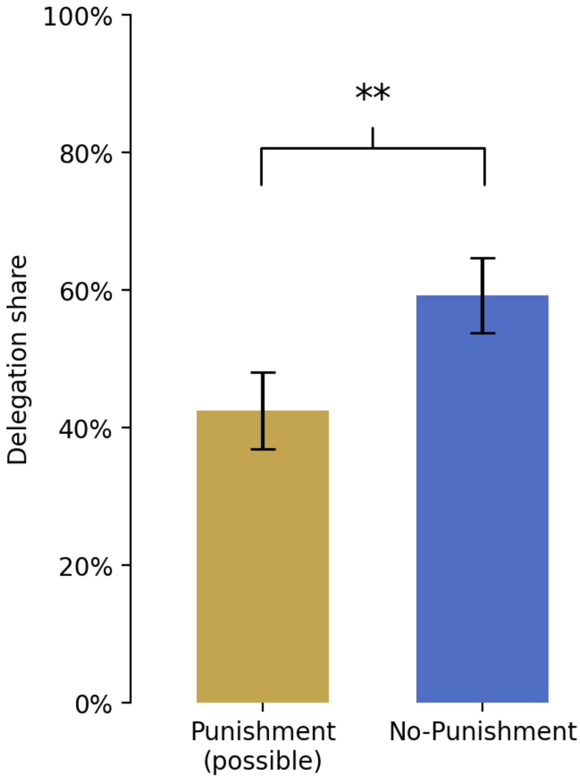
**Result 1.** *The possibility of punishment significantly reduces the likelihood that Player A delegates the decision to the algorithm.*

**[Sub result 2 - Delegation share is around 50%]** Delegation shares varying around 50% indicate that we effectively chose a task that subjects on average credibly perceive as one where their performance and that of the algorithm are comparable.

[13] The preregistration is non-directional on H1; the realized direction is opposite to that anticipated by the responsibility-avoidance literature on which the hypothesis was grounded. We report two-sided tests throughout the body of the paper. The full set of preregistered tests, the deviations from each, and the rationale appears in Appendix **??**.

[14] A within-subject corroborator points the same direction. Player As in the *No-Punishment* condition were asked, after their actual delegation decision, the hypothetical question of whether they would have delegated had punishment been possible. Among the 81 respondents, 12 would have stopped delegating under hypothetical punishment while only 5 would have started, yielding a hypothetical delegation rate of 50.6% (within-subject McNemar's test $p = 0.090$). Hypothetical responses elicited after the actual decision are subject to motivated reasoning; the within-subject pattern complements rather than substitutes for the between-subject test.

Figure 2: Delegation shares by condition




*Note:* Bars show the share of Player As who delegated the final prediction to the algorithm in each between-subject condition. The two-sided difference is significant at the 5% level under both a Pearson Chi-squared test ($p = 0.033$) and a Fisher's exact test ($p = 0.041$). The preregistration does not specify a test direction; the realized effect reverses the sign anticipated by the responsibility-avoidance literature on which the hypothesis was grounded.

That delegation shares vary around 50% indicates that we have effectively chosen a task that subjects on average credibly perceive as one where their performance and that of the algorithm are comparable; we are therefore not picking up an effect that operates only at extreme priors over the algorithm's relative quality.

**[Main result 1 - Regression analysis (logit) - Regression Table - Main result 1 is robust to controlling for socio-demographic characteristics, etc.]** Next, we examine the robustness of the observed treatment difference in the delegation decision. Specifically, we assess whether the difference in delegation rates remains when controlling for socio-demographic characteristics. Table **??** reports the main results from a Logit model. Consistent with the non-parametric test results we find a significant treatment effect on the delegation decision that remains robust to controlling for task performance, socio-demographic characteristics and technology affinity.[15] [Comment: Should we somehow transform the coefficient into something interpretable? Perhaps split this into two specifications - different sets of characteristics - one with performance, one without? Perhaps also add a robustness check, either with the other performance measure or attention check passing.] [Comment: Perhaps include the full table here and discuss some of the other observations? Careful when interpreting the coefficients of covariates as they are quite different in the main treatment regression compared to the regression focussing on the punishment treatment alone.]

**With controls.** Table **??** reports estimates from a Logit model of the delegation decision. Column (1) reports the unconditional treatment effect; Column (2) adds Player A's task performance and a vector of socio-demographic and technology-affinity controls. The treatment effect is robust: the coefficient on the *No-Punishment* indicator is 0.68 ($p = 0.034$, two-sided) in the unconditional specification and 0.76 ($p = 0.023$, two-sided) with controls. The full set of estimated coefficients is reported in Appendix Table **??**. (Column (3), which uses Player A's punishment beliefs and is restricted to the *Punishment* condition, is discussed in Section **??** alongside the belief evidence.)

The paragraph below may have to be integrated with the previous paragraph, or included in the discussion of results.

**Robustness.** The treatment effect survives several further checks. Controlling for Player A's overconfidence (defined as the weighted-average belief about own performance minus actual performance) leaves the *No-Punishment* coefficient essentially unchanged (0.77, $p = 0.022$); overconfidence itself is not significantly associated with delegation ($p = 0.272$). Relaxing the preregistered attention-check exclusion does not change the result either. The full robustness specification is reported in Appendix Table **??**; we report a separate analysis of order effects in Section **??**.

[15] Full regression results are available in Appendix **??**, Table **??**.

Table 1: Determinants of the delegation decision

| | Dependent variable: *Delegation* | | |
|---|---|---|---|
| | *Full sample* | | *Punishment only* |
| | (1) | (2) | (3) |
| No-Punishment indicator | 0.677 | 0.757 | |
| | (0.320) | (0.334) | |
| Punishment beliefs: bad outcome (no del. − del.) | | | 0.937 |
| | | | (0.751) |
| Punishment beliefs: good outcome (no del. − del.) | | | 1.579 |
| | | | (0.741) |
| Task performance | | −0.192 | −0.609 |
| | | (0.128) | (0.236) |
| Controls | no | yes | yes |
| N | 161 | 159 | 60 |
| Pseudo $R^2$ | 0.020 | 0.049 | 0.246 |

*Note:* Coefficients from a Logit regression of the delegation decision on the condition indicator, task performance, and socio-demographic controls. Robust (HC1) standard errors in parentheses. Two-sided $p$-values for the headline coefficients: *No-Punishment indicator* Col (1) $p = 0.034$; Col (2) $p = 0.023$; *Task performance* Col (3) $p = 0.010$; *Punishment beliefs: good outcome* Col (3) $p = 0.033$. Controls in Columns (2) and (3) include age, gender, self-reported socio-economic status, university education, technology affinity, and leadership experience; full results in Appendix Table **??**. Two subjects are dropped in Column (2) due to missing socio-demographic data. Column (3) restricts to the *Punishment* condition and includes within-subject differences in expected punishment between non-delegated and delegated decisions for each outcome; subjects who failed the belief-screen attention check are excluded, as preregistered. Following AEA editorial policy, significance stars are not reported in the table; exact $p$-values are stated in this note.

## Player B – Punishment behavior

**[Sub result 6 - Positive punishment in all scenarios, Punishment higher for bad outcomes than good outcomes]** Next, we evaluate the punishment behavior of Player B in the *Baseline* condition. Throughout the analysis, we exclude subjects that did not pass the attention check that was included on the screen of punishment elicitation.[16] However, all results are robust to including those that fail the attention check [Comment: Maybe do it the other way around? Also, depends on whether we exclude Player A's in the analysis related to punishment beliefs if they failed an attention check]. Punishment in all four scenarios is significantly higher than zero. [Comment: verify that this is statistically correct.] This may be surprising, especially for good outcomes, given that punishment even was costly. However, Player B punishes significantly more for bad outcomes than good outcomes, suggesting that punishment behavior carries information rather than being purely random. 41.8% of Player Bs do not punish in any of the four scenarios. [Comment: Maybe as sanity check: How many subjects are there that forgo 10 cents just to punish little? Perhaps show the distribution of punishment, both across subjects between scenarios, or the difference between scenarios across subjects.] [Coffmann also finds punishment for fair outcomes, and there it is intentional (they could not have been nicer with certainty), whereas in my setup the evaluator can have strong believes as to whether the decision should have been delegated or not and punish accordingly even if the outcome hapens to be good. Maybe mention to show that noise is normal?]

**[Main result 2 - No lower punishment for delegated decisions - Figures]** Moreover, conditional on the outcome for Player B, Player A is not significantly differently punished whether she delegates to the algorithm or made the decision herself (Figure **??**). If Player B earns the high payoff, average punishment coincides between delegation choices (€0.311 when delegating versus €0.310 when not delegating). Similarly, when due to a bad prediction Player B only earns the low payoff, there is only an insignificant tendency towards Player A being punished slightly less when she delegated the decision to an algorithm (€0.494 when delegating versus €0.571 when not delegating, paired t-test p-value: 0.153, Wilcoxon signed-rank test p-value: 0.533). [ This punishment behavior would be inconsistent with delegation decision, no?] The same pattern arises when looking at punishment frequencies (Appendix **??**, Figure **??**). Thus, Player B does not only punish similar amounts for delegated and self-made decisions, but also similarly often. Therefore, in our experimental setting Player A cannot avoid punishment by delegating the decision to the algorithm. Instead, punishment seems to be entirely dependent on the outcome of the prediction.

**Headline.** We now turn to Player B's punishment behavior in the *Punishment* condition. Punishment in all four (delegation, outcome) cells of the strategy method is significantly greater than zero, and Player Bs punish more for low-payoff outcomes than for high-payoff outcomes—

[16]Such an exclusion was pre-registered.

an indication that punishment carries informational content rather than being noise; 41.8% of Player Bs nonetheless do not punish in any of the four scenarios. Conditional on the outcome, however, Player A is not significantly differently punished depending on the delegation decision (Figure **??**). When Player B receives the high payoff, average punishment is £0.311 for delegated decisions and £0.310 for self-made decisions—a difference of one-tenth of a penny. When Player B receives the low payoff, average punishment is £0.494 when Player A delegated versus £0.571 when she made the decision herself; this difference points in the H2 direction but is far from significant (paired $t$-test $p = 0.153$; Wilcoxon signed-rank $p = 0.533$). The same picture obtains when measuring punishment frequencies rather than amounts: Player Bs neither punish delegated decisions more harshly, nor do they punish them more frequently (Appendix Figure **??**). As preregistered, the within-subject test is identified off the 67 Player Bs in the *Punishment* condition who passed the attention check; the result is essentially identical when these subjects are included (mean difference £0.071 vs £0.077; paired $t$-test $p = 0.131$ vs $p = 0.153$; Appendix Table **??**).

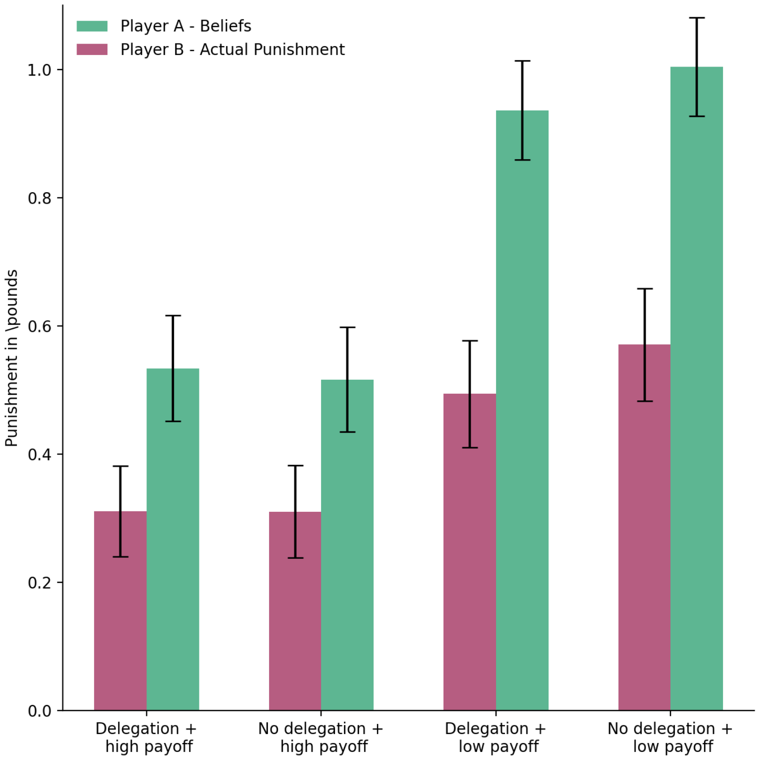
**Result 2:** Player A cannot avoid punishment by delegating the decision to the algorithm.

**Result 2.** *Conditional on the outcome for Player B, we cannot reject the hypothesis that Player A is punished equally for delegated and self-made decisions. A TOST equivalence test rules out moderate insulation effects (those exceeding* £0.20*, or* 10% *of the maximum punishment) at* $\alpha = 0.05$*, but we cannot rule out smaller insulation effects.*

**Equivalence (TOST).** We can refine the negative result with a positive equivalence claim. A two one-sided test (TOST) at the ±£0.20 bound—against the null that the true insulation effect exceeds 10% of the maximum punishment—rejects in both the bad-outcome cell ($p = 0.012$) and the good-outcome cell ($p < 0.001$). At a tighter ±£0.10 bound, equivalence cannot be established ($p = 0.332$ for the bad-outcome cell). In our setting we can therefore rule out moderate insulation effects (above £0.20 per cell) with high confidence, but cannot rule out small ones below this bound.

**What predicts punishment?** Appendix Table **??** reports a regression of chosen punishment on a delegation indicator, an outcome indicator, their interaction, Player B's beliefs about Player A's performance, and a vector of socio-demographic controls. The interaction term is small and statistically insignificant ($p = 0.137$), confirming the bivariate result. Punishment is also independent of Player B's belief about Player A's performance: across all four cells, Pearson correlations between cell-specific chosen punishment and Player B's weighted-average belief about Player A's score are essentially zero ($|r| \leq 0.03$, all $p > 0.8$). This rules out the alternative explanation that Player Bs who think Player A is more capable systematically punish less (or more); punishment is not driven by performance-conditional inference about

Figure 3: Punishment by Player B and beliefs about punishment by Player A




*Note:* Average punishment chosen by Player B (solid bars) and average belief about punishment held by Player A (hollow bars) for each of the four scenarios. Bars are reported in pounds and standard errors are reported as error bars. Beliefs are elicited prior to learning the realized outcome and on the same scale as the punishment decision; the belief overlay is analysed in Section **??**.

Player A.[17]

## 4.1 Discussion of results

**[Discussion of Main result 1 - Introduction]** The observed decrease in the likelihood to delegate following the introduction of potential punishment can be interpreted in several ways.

**[Discussion of Main result 1 - Different setup to past studies: incentives aligned, inherent uncertainty - Based on: main result 1 - Links to: Punishment beliefs]** Thus, we not only reject *Hypothesis 1* but also observe an effect in the opposite direction, suggesting that the possibility of punishment discourages delegation to the algorithm. Our hypothesis was grounded in previous empirical evidence, though these studies were conducted in settings that differ from ours in various important ways. Most notably, our task departs fundamentally from the modified dictator games used in **????**. In such games, the decision intent is more transparent: participants make either fair or unfair allocations that are directly payoff-relevant to themselves and implemented with certainty. In contrast, our prediction task involves uncertainty (subjects may err even though they put in substantial effort) and the incentives of Player A and Player B are aligned.[18] These aspects may obfuscate the intent of Player A, possibly diluting an intention-based punishment motive for Player B [Comment: "...and therefore the need to avoid punishment by Player A"] [This would (i) be reflected in punishment beliefs, and (ii) not explain the reversal, but merely the absence of any treatment effect]. Nevertheless, we chose a prediction task because prediction under uncertainty is precisely the domain where algorithmic decision-making is most prevalent.

The observed decrease in the likelihood to delegate following the introduction of potential punishment can be interpreted in several ways. We not only fail to support *Hypothesis 1*; the realized effect points in the opposite direction, suggesting that the possibility of punishment discourages delegation to the algorithm. Because the reversal was not preregistered, the analyses in this subsection are exploratory and we treat the supported mechanisms as candidate hypotheses rather than as established. We consider four candidates—anticipated differential punishment (M1), effort to outperform the algorithm (M2), process ownership (M3), and experimenter demand (M4)—and weigh each against the data, taking care to distinguish what we can rule out from what we can only flag as consistent.

[17] Post-experiment Likert measures provide a tentative attitudes–behavior link. Among the 44 Player Bs who passed the attention check and completed the post-experiment questions, those who reported that Player A should feel more responsible for the decision punish self-made good outcomes more relative to delegated good outcomes (Pearson $r = 0.36$, $p = 0.015$ between the responsibility Likert and the within-subject difference $punish\_nodel\_good - punish\_del\_good$); the corresponding pattern in the bad-outcome cell goes in the same direction but is not significant ($r = 0.24$, $p = 0.114$). When asked directly who should be accountable for AI-generated decisions, 66% of these subjects identified deploying companies as primarily accountable, 18% developers, 14% end-users, and only one named government or regulatory bodies. The sub-sample is small ($n = 44$); we treat these patterns as suggestive.

[18] We consider the incentives of Player A and B to be aligned, even though Player A's payoff is only indirectly affected by her own decision, insofar as a better outcome for Player B reduces expected punishment.

**Departures from prior settings.** Our hypothesis was grounded in previous empirical evidence, though these studies were conducted in settings that differ from ours in important ways. Most notably, our task departs fundamentally from the modified dictator games used in **????**. In such games, the decision intent is more transparent: participants make either fair or unfair allocations that are directly payoff-relevant to themselves and implemented with certainty. In contrast, our prediction task involves uncertainty (subjects may err even though they put in substantial effort) and the incentives of Player A and Player B are aligned.[19] These aspects may obfuscate the intent of Player A, possibly diluting an intention-based punishment motive for Player B. The same reading also helps interpret Result 2: because the algorithm is calibrated at the individual level and this calibration is common knowledge, the delegation decision in our setting carries no signal about Player A's belief in her own ability relative to the algorithm. Player B has no obvious basis on which to read the delegation decision as overconfident, lazy, or selfish, which is exactly the channel through which delegation might insulate from punishment in less-calibrated settings (as in **?**). We chose the prediction task nevertheless because prediction under uncertainty is precisely the domain where algorithmic decision-making is most prevalent.

**[Discussion of Main result 1 - Punishment beliefs as potential channel: Stronger anticipated punishment for delegated decisions, either due to (i) morality of the decision, or (ii) punishment for shirking - Based on: main result 1 - Links to: Punishment beliefs]** First, Player A may anticipate to receive stronger punishment for a delegated decision compared to a self-made decision. This belief pattern would be consistent with parts of prior work, suggesting that observers view delegating moral decisions to algorithms relatively critically. [Gogoll Uhl 2018, Goldbach et al 2019, Niszczota Kaszas 2020, Jauernig et al 2022]. Alternatively to being driven by the morality of the decision, the same belief pattern would also be consistent with a simple shirking motive. In this case, Player A would expect greater punishment for shirking their effort to make a decision for Player B, but, in the absence of punishment, could shirk without monetary consequences. In both cases, delegation behavior should be consistent with observed beliefs about punishment. [Fore-shadowing not well-placed -¿] However, as will be discussed in [refer to discussion of beliefs], we do not observe that subjects expect harsher punishment for a delegated decision.[Comment: Perhaps revers all these arguments: Not that this anticipated punishment pattern is consistent with these motives, but rather that all these motives would prescribe such a punishment.]

- **[Shirking - Link to Player B punishment behavior]** Player B punishment is outcome-based, not delegation-based, which is not surprising given that performance is controlled for. However, if shirking mattered we could think about observing more punishment when delegating, even for the good outcome. Hence, this can explain positive punishment at least for good outcomes when delegating.

[19] We consider the incentives of Player A and B to be aligned, even though Player A's payoff is only indirectly affected by her own decision, insofar as a better outcome for Player B reduces expected punishment.

**[Punishment Beliefs] [Sub result 7 - Player A overestimates punishment - Based on: Punishment beliefs]** Average punishment of Player B is significantly overestimated by Player A (Figure **??**). [Comment: What about punishment conditional on Player A punishing in any scenario? But: If we exclude Player B that never punishes, we would also have to compare to player A beliefs excluding those that antcipate to never be punished.] However, relative punishment patterns are on average correctly anticipated, i.e. that conditional on the outcome punishment does not differ between delegated and non-delegated decisions. Therefore, neither does Player 2 punish delegated decisions more or less, nor do Player A's beliefs suggest that she anticipates Player B to punish delegated decisions more or less strongly. [Comment: This goes back again to Punishment beliefs of Player A and convolutes the analysis here.]

**Punishment beliefs as a candidate channel.** A natural reading of Result 1 is that Player A anticipates stronger punishment for a delegated decision than for a self-made decision and delegates less in response. This belief pattern would be consistent with parts of prior work suggesting that observers view delegating moral decisions to algorithms relatively critically (e.g., **???**). Alternatively, the same belief pattern would be consistent with a simple shirking motive, in which Player A expects greater punishment for shirking the effort to make a decision for Player B but, in the absence of punishment, can shirk without monetary consequences. In either case, delegation behavior should be consistent with observed beliefs about punishment. As we now show, the belief evidence does not support this pattern.

**[Sub result 5 - Punishment beliefs: Beliefs partly hypothetical in Baseline and purely hypothetical in Treatment, Construct belief measures as differences, Regression table]** After the delegation decision, we elicit Player A's beliefs about Player B's punishment behavior. First, we check the consistency of punishment beliefs with the delegation decision for the Baseline condition only, because this is where actual punishment is possible, whereas elicited beliefs in the treatment condition concern a purely hypothetical scenario. Thus beliefs may not be comparable across treatments. [Introduce beliefs as an incredibly noisy measure: They are unincentivized, because of hedging. In Baseline, two scenarios are hypothetical, in Treatment, all scenarios are hypothetical. Check whether beliefs vary with the delegation decision. "Of course beliefs were not incentivized and potentially motivated. We don't know whether they delegate or not because they hold these specific beliefs about punishment or they hold these specific beliefs since they delegated."] Specifically, we employ two measures that compare the difference in expected punishment of a self-made and delegated decision for each of the two possible outcomes for Player B, so that higher values of these two measures correspond to higher expected punishment for delegated decisions relative to non-delegated decisions. Player As that, conditional on the outcome for Player B, expect to be punished more when making the decision for Player B themselves rather than delegating, tend to delegate the decision to the algorithm (Table **??**, column 3). [Comment: Maybe we can also have a joint measure by adding up both differences and only afterwards disentangeling them?] Thus, even though the belief measure was not incentivized, and even though at the time of the belief elicitation, two of the four scenarios could not materialize anymore, the belief measure carries some informational value.[20] [Comment: Maybe move to discussion, because this is about interpretening the treatment difference?] [Comment: Additional analysis: Are beliefs correlated with delegation decision? Does everyone feel the same or is there large heterogeneity? Maybe show belief distributions for the four scenarios. Compare beliefs for delegated decisions (both outcomes) between those that delegated and those that didnt in Baseline (hypothetical vs. relevant)]

**[Sub result 1 - regression analysis (logit) - negative correlation between performance and delegation when punishment is possible]** The regression framework let's us also explore, which other factors may influence the delegation decision. Interestingly, in the *Baseline* condition, Player A's performance is negatively correlated with the likelihood to delegate (Pearson correlation coefficient = -0.19 (p-value: 0.092)), whereas this relationship is much weaker in the *Treatment* condition, rendering the regression coefficient on task performance insignificant without treatment interaction. [Since this only becomes obvious from the regression for the Punishment Treatment subsample alone, we can use that as a starting point.] [Comment: Rhetorically, we start with a test here, and not with the coefficient we observe. Should we immediately interact performance with treatment, if that even helps to produce a significant result?] [Comment: We can plot the share of people that delegated on y-axis and the overall score on the x-axis. For the Baseline condition, we will see a falling trend whereas for the treatment condition we see a u-shape. Make the dots reflect the number of people with these scores.] [Comment: Feier et al. - Here it makes sense that the worse you are, the more likely you delegate, because it increases expected success for the third party. But in my experiment this is not clear and therefore more interestuing. Maybe point this out?] This suggests that when punishment is possible, better-performing Player As are less inclined to delegate the decision, despite common knowledge about its equal performance.[21] The higher their likelihood of securing the high payoff for Player B, the more inclined Player A is to having directly brought about the outcome. [Comment: On performance, from Steffel et al: "When a decision maker chooses for someone else in a scenario with only positive anticipated outcomes, the individual may not mind having or may even prefer to maintain decision control, as they could potentially reap any rewards associated with making a good decision (Bartling Fischbacher, 2012). However, when faced with choosing on behalf of another person in a situation with only negative potential outcomes or with the possibility of a negative outcome, decision makers may prefer to delegate to someone else in fear of receiving the punishment or blame associated with making a bad decision. Decision makers may expect that delegation will help them avoid blame from both themselves and from others. [...] Experiment 3 demonstrates that both anticipated responsibility and blame contribute to the tendency for people to delegate potentially costly choices for others and that people care more about avoiding blame for bad outcomes than getting credit for good outcomes.]

[20] As pre-registered, we reduce noise in belief elicitation by excluding subjects that fail to pass the attention check presented on the same screen as the belief elicitation. [Comment: What if we do not do this?]

[21] Similarly, the delegation decision of Player A is negatively related to their beliefs about own performance, even though not significantly so.

**[Discussion of Sub result 1 - Based on: punishment beliefs]**

- **[Punishment beliefs - two channels]** Finding this relationship in the *Baseline* condition only indicates that Player A does not merely avoid directly causing a bad outcome and savoring directly causing a good outcome for Player B. Instead, it suggests a connection to anticipated punishment, such that: [Think more about this potential non-linearity, i.e. If I know it will fail, then I may want to delegate. If I know I will succeed, I may want to do it myself. Hence, my preference on delegation may change depending on the chance of success. What kind of literature is here? This will also relate to the point about the amount of inherent uncertainty in the decision.]
  - **[Channel I: Diff-in-Diff between punishment beliefs]** Player A expects more severe punishment for delegated decisions (relative to non-delegated ones) when the outcome is good than when it is bad. We do not find this pattern in punishment beliefs. [Insert evidence; Is this statement even correct?]
  - **[Channel II: Punishment beliefs correlate with task performance]** Alternatively, better performing Player As could hold systematically different punishment beliefs than worse-performing Player As. However, we find no systematic differences in expected punishment patterns depending on the performance of Player A. [How are beliefs connected to punishment beliefs? Do those Player As that perform better anticipate to be punished less? Does performance relate to the diff-in-diff in beliefs?]

- **[Inherent uncertainty]** Bharti and Dietvorst provide suggestive experimental evidence on how the amount of uncertainty inherent in a decision affects the preference for algorithms. They find that with increasing task uncertainty, subjects' preference for algorithms over their own prediction decreases, even though they were aware that the algorithm performs better on average. [But worse performance does not equal more uncertainty. I can be certain that I lose. But we could assume that uncertainty is maximized at a score of 50%, and we approach that performance from the bottom. So better performance comes with greater outcome uncertainty and greater uncertainty is linked to making the decision yourself. Again, this reason would apply to treatment and baseline equally.] We find this in a setting in which the decision is payoff-relevant to another participant, equal relative performance is common knowledge and punishment is possible. [Key question, with higher performance comes lower or higher uncertainty? Depending on the answer to this question [check Bharti paper], our finding is in cotrast or in line with the literature]

**Punishment beliefs.** After making the delegation decision but before learning the outcome, we elicited Player A's beliefs about Player B's punishment in each of the four scenarios. Two patterns emerge. First, Player As substantially *overestimate* the level of punishment they will face: average expected punishment exceeds average realized punishment in every scenario (Figure **??**, hollow bars). Second, however, Player As correctly anticipate the *relative* pattern of punishment: they expect no significant difference in punishment between delegated and non-delegated decisions, conditional on outcome—a pattern that closely tracks the actual punishment behavior of Player B documented in Result 2. *It is this relative-pattern finding—on punishment beliefs alone—that rules out a simple anticipation mechanism for Result 1.* If Player A had reduced delegation in the *Punishment* condition because she anticipated being punished more harshly for delegation than for self-made decisions, we would expect to see this asymmetry in the elicited beliefs. We do not.

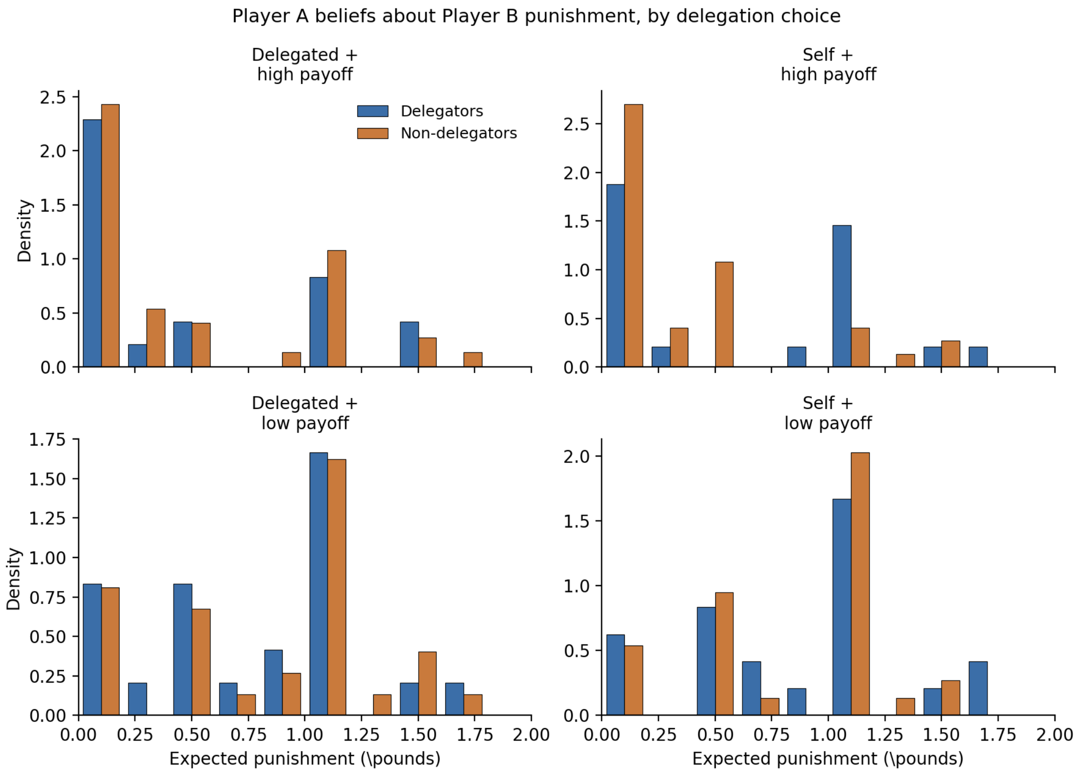
Despite the weakness of the elicitation, punishment beliefs carry within-subject information. Column (3) of Table **??** reports a Logit regression of delegation on Player A's belief difference (*no delegation* minus *delegation*) for each outcome, restricted to the *Punishment* condition.[22] Coefficients on both belief differences are positive: a Player A who expects to be punished more for non-delegation than for delegation in a given outcome is more likely to delegate. The coefficient on the good-outcome belief difference is significant at conventional levels (1.58, two-sided $p = 0.033$); the coefficient on the bad-outcome belief difference is positive but noisier (0.94, two-sided $p = 0.21$). The within-subject signal is asymmetric across cells (Figure **??**): the single significant difference between delegators and non-delegators is in the *self decision + good outcome* cell, where delegators expect punishment of £0.82 while non-delegators expect only £0.32 ($t$-test $p = 0.006$, Mann-Whitney $p = 0.019$); for the other three cells the gap is small and not significant. Because beliefs were elicited after the delegation decision and the elicitation was unincentivised, we cannot rule out that subjects rationalised their choice by holding consistent post-hoc beliefs; the within-subject correlation is consistent with the standard interpretation that anticipated punishment shapes delegation, but does not identify the causal direction.

**(M1) Anticipated differential punishment.** Bringing the belief evidence to bear on the four candidate mechanisms in turn: a Player A who expects to be punished more harshly for a delegated decision than for a self-made decision will rationally delegate less when punishment is on the table. The belief evidence above shows that Player As do not on average anticipate harsher punishment for delegated decisions, and Player B does not in fact deliver such punishment (Result 2). M1 is therefore inconsistent with both the elicited beliefs and the realized punishment, and we view this candidate as ruled out by the available evidence.

**(M3) Process ownership of the outcome.** A third mechanism, suggested by the heterogeneity-by-performance pattern documented above, is that Player As who have a higher subjective probability of producing the high payoff for Player B prefer to bring about that good outcome

[22] The two belief measures compare expected punishment of a self-made and delegated decision for each of the two possible outcomes for Player B, so that higher values correspond to higher expected punishment for delegated decisions relative to non-delegated decisions. Subjects that failed the attention check on the belief screen are excluded, as preregistered.

Figure 4: Player A beliefs about Player B punishment, by delegation choice




*Note:* Distribution of Player A's expected punishment from Player B in each of the four (delegation, outcome) cells, separately for Player As who actually delegated (blue) and those who did not (orange). *Punishment* condition, attention-check passers ($n_{\text{deleg}} = 24$, $n_{\text{nodeleg}} = 37$). The only cell with a statistically significant difference is *self decision + good outcome* (top right): delegators expected £0.82 vs non-delegators £0.32 ($t$-test $p = 0.006$).

themselves rather than via the algorithm. The intuition draws on the psychological-ownership literature: people derive utility from being the proximate cause of a good outcome for another, and this utility is asymmetric with respect to outcomes (?). The data are consistent with this mechanism, but the supporting evidence is thin. The negative relationship between performance and delegation in the *Punishment* condition is what process-ownership predicts: better-performing Player As, who are more likely to deliver the high payoff, prefer to do so personally. The absence of this relationship in the *No-Punishment* condition is consistent with process ownership being amplified by the moral salience that the punishment possibility introduces. We are explicit, however, that the supporting evidence consists of a single heterogeneity pattern in a 60-subject sub-sample and a comparison of selected non-delegator success rates; the pattern is the most consistent reading of the data we have, but not a confirmed mechanism. A direct test would require a treatment that varies process ownership while holding the punishment possibility fixed—for example, an arm in which the algorithm is presented as Player A's own "tool" rather than as a separate agent.

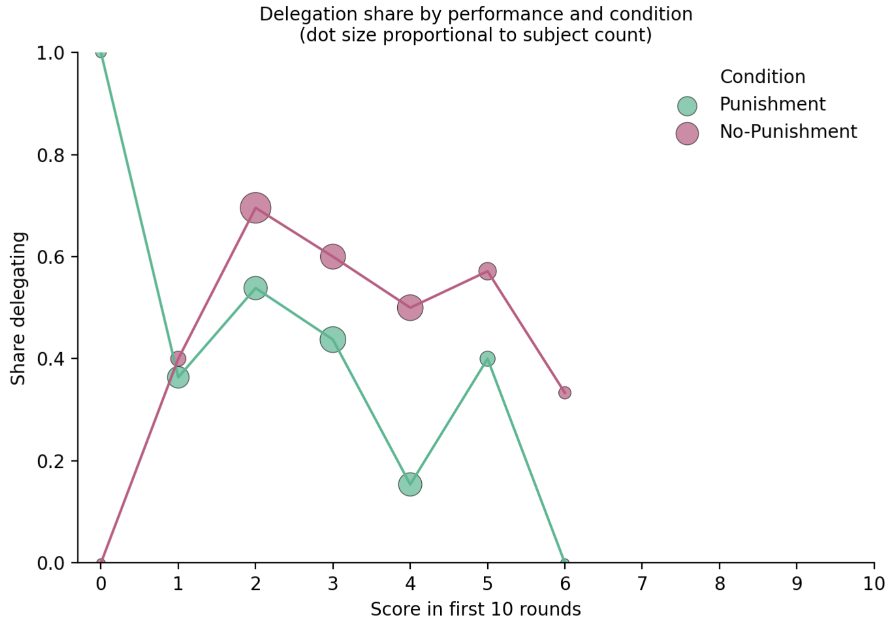
[**Performance differences - not responsible for treatment difference**]

- [**Sub result 3 - Performance and main result 1 robustness: Treatment differences in delegation are not due to performance differences in first 10 tasks - Figure**] The average subject scores a point in three out of the first ten rounds and performance is balanced across treatments (Figure **??**). Thus, treatment differences in the propensity to delegate do neither stem from ex-ante differences in performance between both treatments, nor, as shown before/will be shown, from increased effort for the decision for Player B in the Baseline condition. [Comment: The order will most likely change.]
- [**Sub result 4 - Overconfidence - Figure**] Moreover, subjects exhibit significant overconfidence, measured by the weighted average belief about their performance. While this overconfidence may also partly reflect the information that the algorithm is performing equally well, it is not statistically different between both treatments. [Confidence in their performance may be more important for the delegation decision, so we should talk about how they perceived their performance.] [is the degree of overconfidence predictive for anything? Can we not measure overconfidence as a binary variable maybe as a robustness check?]

[Comment: Include in the figure the split by who ended up delegating?]

**Heterogeneity by performance.** The regression framework also lets us explore which other factors influence the delegation decision. Within the *Punishment* condition, Player As who performed better on the first ten rounds were less likely to delegate. The Pearson correlation between the score in the first ten rounds and the probability of delegation is $-0.19$ (two-sided $p = 0.092$); the corresponding regression coefficient in Column (3) of Table **??** is $-0.61$ (two-sided $p = 0.010$). The strengthening of the relationship in the regression reflects, in part, the attention-check restriction (Column (3) is identified off $n = 60$ Punishment-attention-pass subjects, while the raw correlation uses all 80 Punishment subjects). In the *No-Punishment* condition, no comparable relationship between performance and delegation is observed. Figure **??** visualises this asymmetry. The pattern indicates that Player As do not merely avoid directly causing a bad outcome (or savor causing a good one); rather, the higher their likelihood of producing the high payoff for Player B, the more inclined they are to bring about that outcome themselves.[23] This becomes the empirical hint that motivates the process-ownership reading developed below.

Figure 5: Delegation share by performance and condition




*Note:* Each point reports the share of Player As at the corresponding first-ten-rounds score who delegated the final prediction. Dot size is proportional to the number of subjects at that score. The downward slope in the *Punishment* condition is the Pearson $r = -0.19$ ($p = 0.092$); no comparable slope in *No-Punishment*. Restricted to attention-check passers.

Two routes through which a performance-delegation relationship in the *Punishment* condition could in principle be reconciled with anticipated punishment are (i) a diff-in-diff in expected punishment between the good and bad outcome cells, conditional on delegation, and (ii) systematic differences in punishment beliefs by performance. Neither is borne out: we do not find evidence that Player As expect a relatively larger punishment penalty for delegation in good-outcome states than in bad-outcome states, and punishment beliefs do not vary systematically with Player A's performance.

**[Delete?]**

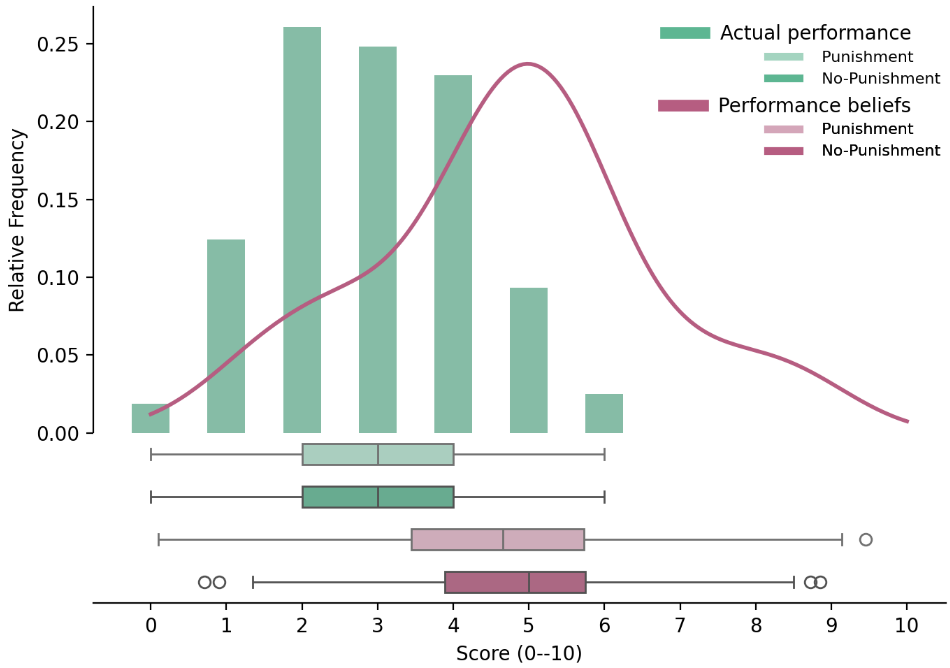
**Performance beliefs.** As a sanity check on the elicitation, we note that both players overestimate Player A's task performance. The true mean overall score in the first ten rounds is

[23] Similarly, the delegation decision of Player A is negatively related to her own beliefs about her performance, even though not significantly so.

2.93/10, while Player A's weighted-average belief about her own score is 4.76 (overestimate of 1.84 score points) and Player B's weighted-average belief about Player A's score is 4.32 (overestimate of 1.40 score points). The overestimation is therefore not a Player-A-specific or Player-B-specific artefact, but a feature of the elicitation. Performance beliefs do not enter the M1 ruling-out argument; we report them here as a sanity check rather than as evidence on the mechanism.

**Performance differences and overconfidence across conditions.** Treatment differences in delegation are not driven by performance differences across conditions in the first ten rounds. The average subject scores correctly in roughly three of the first ten rounds and performance is balanced across conditions (Figure **??**). Subjects also exhibit significant overconfidence, measured by the weighted-average belief about own performance; while this overconfidence may partly reflect the information that the algorithm is performing equally well, it is not statistically different across conditions and—as noted in the Robustness paragraph above—is not significantly associated with delegation.

Figure 6: Performance in the first ten rounds, by treatment




*Note:* Distribution of correct predictions (within 10 lbs of the true weight) over the first ten task rounds, by condition. Average performance is 2.93 correct out of ten and is statistically indistinguishable across conditions.

**[Discussion of Main result 1 - Higher effort for decision for Player B, making delegation less attractive - Based on main result 1 - Links to performance and performance beliefs, but not necessarily the relationship between performance and delegation]** Second, although our design controls for relative performance between Player A and the algorithm based on the first ten rounds, potential punishment may motivate Player A to exert greater effort in making the decision for Player B, thereby choosing not to delegate in hopes of outperforming the algorithm. [Comment: If effort increased in both conditions: effort exertion would be altruistic, and would be treatment-independent. If effort only increases with punishment, then the motive would be driven by the expectation to decrease the likelihood of punishment.] [In this whole section, it is about actual performance, not beliefs about performance, because performance is a proxy for effort.]

- Yet, this mechanism is not supported by the data.
- **[Analysis of potential mechanism - If punishment is possible, performance was actually lower, not higher.]** Among Player As who chose not to delegate, those in the *Treatment* condition (where punishment is not possible and the delegation share is higher) significantly outperformed those in the *Baseline* condition. Specifically, 45.5% of non-delegating Player As in the *Treatment* condition earned the high payoff for Player B, compared to only 21.7% in the *Baseline* condition. [Comment: Maybe add a sentence that performance can be seen as a proxy for effort.]
- **[Analysis of potential mechanism - Performance difference in decision for Player B not driven by pre-existing skill gap.]** This difference is not driven by pre-existing performance differences across the first ten rounds.[24] [Comment: How do we go about no performance differences in first 10 rounds between the Player As that delegated in both treatments, the delegation share being higher if punishment is not possible and delegation decreasing with performance (in both or only one condition)?]
- Therefore, if anything, increased effort appears to be concentrated among Player As in the *Treatment* condition, where punishment is not possible. [Comment: Therefore, we see no evidence for this explanation.]

**(M2) Effort exertion to outperform the algorithm.** An alternative mechanism is that the prospect of punishment motivates Player A to exert greater effort on the final prediction, in the hope of personally producing the high payoff and thereby choosing not to delegate to outperform the algorithm. Although our design controls for relative performance based on the first ten rounds—making the algorithm and Player A equivalent in expectation on the final prediction—potential punishment could in principle shift Player A's effort margin. The data are inconsistent with the simple form of this story. Among Player As who chose not to delegate, those in the *No-Punishment* condition (where punishment is not possible and the

[24] Performance across the first 10 rounds does not differ significantly between Player As that did not delegate in the *Treatment* (3.12/10) and *Baseline* (2.97/10) conditions. Measuring performance improvement as the difference between the average absolute prediction error across the first ten rounds and the absolute prediction error in the decision for Player B, we find that Player A in the *Treatment* condition significantly improves their performance, whereas Player A in the *Baseline* condition performs as before. [STATA: both improve, but more in Treatment] Although this performance increase may not necessarily stem from higher effort, it could also not be due to task-specific variation in difficulty, as all Player As across both treatments were presented with the same images in the prediction task.

delegation share is higher) outperformed those in the *Punishment* condition: 45.5% of non-delegating Player As in the *No-Punishment* condition earned the high payoff for Player B, compared to 21.7% in the *Punishment* condition. This difference is not driven by ex-ante performance differences across the first ten rounds (*No-Punishment*: 3.12/10; *Punishment*: 2.98/10; two-sided $p = 0.66$); the cross-condition mean score in the first ten rounds is 2.93/10 overall and statistically indistinguishable across conditions (Figure **??**), so the M2-relevant gap among non-delegators reflects selection on delegation rather than ex-ante condition differences.[25] Counter-intuitively, increased effort thus appears concentrated among non-delegators in the *No-Punishment* condition, where punishment is *not* on the table—incompatible with a story in which Player A in the *Punishment* condition tries especially hard to outperform the algorithm in order to avoid punishment.

**(M4) Experimenter demand.** A final possibility is that the punishment possibility cues Player A to behave more conservatively—for instance, to avoid an action (delegation) that the experimental setting itself implicitly frames as morally questionable. We cannot rule this mechanism out with our data. The most we can say is that two implications of a strong demand story are not borne out: if demand were dominant, Player A's behavior should track elicited beliefs more tightly than it does, and the sharp performance heterogeneity in the *Punishment* condition (better performers delegating less) would be hard to explain. Neither absence is decisive against the demand reading. We discuss the implications, and the design moves that would identify demand directly, in Section **??**.

In sum: M1 (anticipated differential punishment) is inconsistent with the data. M2 (effort to outperform the algorithm) is inconsistent with the data in its simple form. M3 (process ownership) is consistent with the data but supported only by exploratory heterogeneity analysis. M4 (experimenter demand) cannot be separated from M3 with the present design. The most economical reading—the one we adopt with explicit hedges in the Conclusion—is that the morally charged context introduced by the punishment possibility makes Player A want to be the proximate cause of any good outcome for Player B, while we acknowledge that an experimenter-demand story can produce qualitatively similar patterns. Disentangling the two is the cleanest direction for follow-up work.

[25] Measuring performance change as the difference between the average absolute prediction error across the first ten rounds and the absolute prediction error on the final prediction, we find that Player As in the *No-Punishment* condition improve their performance from rounds 1–10 to round 11, whereas Player As in the *Punishment* condition do not improve. Because all Player As across both conditions faced the same image on the final prediction, this differential improvement cannot be attributed to task-specific variation. We caveat that the comparison is between selected non-delegator subsamples, not between random subsets of all Player As; selection on the very outcome we are measuring is a known limit of this comparison.

## 4.2 Other

**Order effects.** The delegation decision and the punishment / belief-elicitation screens presented their options in randomised (or block-wise randomised) order. Order effects are localised to the *Punishment* condition: presenting the algorithm option first roughly doubles delegation (52.5% vs 32.5% when the own-decision option is first; $\chi^2$ $p = 0.070$), while the *No-Punishment* condition shows no order effect ($p = 0.674$); the treatment $\times$ order interaction is not significant ($p = 0.115$), so the H1 finding is not driven by interaction with order.

**Power.**

Add an ex-post (or, preferably, the original ex-ante) power analysis here, framed as in **?**.

**[To resolve:]**

- For two subjects, we must be missing realizations of control variables or something, because the sample size decreases from 161 to 159. Do we leave this unaddressed or should we include a footnote?
- Inequality/deservingness motive: Player A may think that Player B does not have to work to potentially earn 5 pounds and consider this unfair. However, Player A does not know what else Player B did have to do. Player B may want to bring down Player A's payoff so that they earn similarly, but also here there is uncertainty and this would just be outcome-based and not prescribe a treatment effect [Comment: Do not discuss this.]
- Is there any benefit in looking at hypothetical punishment of the No-Punishment treatment? Given that there are a lot of hypotheticals also in the Punishment treatment, should we perhaps not even pool for any analysis that only looks at anticipated punishment, however not when relating the delegation decision to anticipated punishment [Comment: Decause if it is entirely hypothetical, why would the beliefs matter for the delegation decision?]; How can we exploit that at the belief elicitation stage, a subset of scenarios are already for sure not relevant, e.g. we could check whether the difference between good and bad for delegated decisions depends on whether the decision was delegated, and the same for non-delegated decisions.
- Similarly to Player A, Player B overestimates Player A's task performance. Therefore, the likelihood with which the good outcome scenario marterializes is anticipated to be higher than it actually is. Can we do anything with this? [Comment: If Player B thinks the task is easy, they could punish more for bad outcomes. Punishment (all four individually, but also differences) is independent (Pearson and Kendall) of beliefs about Player A's performance, so this holds consistently across the whole (mostly supported between 2 and 6) distribution of performance beliefs. Hence, from a punishment perspective it doesn't matter whether you delegate when task is easy or harder. For those thinking longer about

the punishment decision, there is a positive correlation between easiness of the task and punishment for a bad outcome when you don't delegate vs. when you delegate.] Of course, the perception is again tainted by Player B being told that performance was the same.

- What can we do with the hypothetical delegation choice of subjects in the No-Punishment treatment? Empirically, the result moves into the same direction (ttest p-value: 0.089)]
- Are risk attitudes related to any decisions? More risk-seeking leading to less delegation maybe or what could be the hypothesis? [Comment: Could be interpreted as a coefficient as part of the main result 1 regression.] More risk-seeking people could also risk being punished more, but I do not see how that would relate to the results. [Comment: Result in Feier et al.: risk averse subjects were also more likely to delegate.]
- Do robustness analysis with performance not in points, but in terms of distance from the truth.
- There are hedging motives in the performance elicitation task, but let's ignore those.
- Ways to filter: (i) failed attention checks, (ii) did not spend a lot of time on the page (delegation page and beliefs about punishment/punishment page)
- Player B: Is punishment correlated with any socio-demographics. Shoud we not do a regression here as well? Related to this: For those that state that individuals should be held responsible for AI-generated decisions responsibility avoidance is less possible (pos. correlation between 'nodel-del-bad' and 'responsibility') (maybe should recode variable 'responsibility').
- Main result 1 is consistent with Gogoll and Uhl 2018: Algorithm use was punsihed more.
- Check: In python, use ttest$_r elinsteadofttest_i ndtocomparepunishment$.
- Check: Feier et al. use GLM Logit IRLS. Is that different from what I am using?
- Maybe check if Player As that are better/more confident in last guess have different beliefs.
- Scrutinize Feier et al. analysis, e.g. They have the same roles: Is punishment behavior independent of delegation behavior? Is it in line with delegation behavior?
- When showing balance across observables, then be careful to use chi-squared test for categorical variables.
- Show that randomization also worked on time taken, etc?

[Design issue: Why not have punsihment beliefs before delegation decision?] [Design question: Would it have been interesting to compare the delegation behavior between a decision for oneself vs. someone else?]

[Consideration for literature review: Regarding Feier et al: If I always say that punishment in their experiment could be due to the signal value in delegation (concerns regarding relative performance), then this holds for both their treatments so should technically not explain why they find differential rewarding in the machine treatment.]

# 5 Discussion and Conclusion

This paper asks two questions about how the prospect of punishment interacts with the use of algorithms as decision aids. Can decision-makers shift responsibility for bad outcomes onto algorithms by delegating their decisions? And does the anticipation of being held responsible motivate delegation in the first place? We address both questions in a preregistered online experiment in which Player A makes a payoff-relevant prediction for Player B and may delegate the prediction to an algorithm calibrated to perform as well as Player A herself. A between-subject manipulation switches off punishment by Player B in the *No-Punishment* condition; the *Punishment* preserves the natural setting in which the prospect of punishment is on the table.

Two findings emerge. First, and most striking, the *prospect* of punishment *reduces*, rather than increases, the propensity to delegate to the algorithm: 42.5% of Player As delegate when punishment is possible, compared to 59.3% when it is not. This direction is the opposite of what the responsibility-avoidance literature would predict, and we view the reversal as the paper's main empirical contribution—noting that, because the result reverses the preregistered direction, the analyses we use to interpret it (Section **??**) are exploratory rather than confirmatory. Second, conditional on the outcome for Player B, we cannot reject the hypothesis that delegated and self-made decisions are punished equally. The point estimate runs in the predicted direction (lower punishment under delegation in the bad-outcome cell) but is small relative to the test's confidence interval at the achieved sample size; we therefore frame Result 2 as a failure to detect insulation rather than as positive evidence that delegation cannot insulate. Player A does not appear to be able to use this particular algorithm as a meaningful blame shield in the bounded sense the test can assess.

**Mechanism.** Why does punishment reduce, rather than increase, delegation? The data are inconsistent with a simple anticipation mechanism: Player As do not on average expect harsher punishment for delegated decisions, and they correctly anticipate the absence of differential punishment that we observe in Player B's actual choices. The data are likewise inconsistent with the simple form of an effort story in which the prospect of punishment motivates Player A to outperform the algorithm: among non-delegators, those in the *No-Punishment* condition (where punishment is impossible) outperform those in the *Punishment* condition, the reverse of what such a story implies. The reading most consistent with the data is one of *process ownership*: Player As who actively choose not to delegate take responsibility for the outcome, exert correspondingly higher effort, and that share is higher when punishment is on the table. The supporting evidence is exploratory rather than confirmatory, however, and rests on a single performance-by-delegation heterogeneity pattern in a 60-subject sub-sample (Column (3) of Table **??**); we cannot, with this design, separate process ownership cleanly from an experimenter-demand reading in which the punishment cue itself nudges Player A toward direct engagement. We therefore present process ownership as the most consistent reading of the data we have, not as a confirmed mechanism. Disentangling it from demand is a clean direction for follow-up work, sketched below.

This refinement is consistent with, but goes beyond, the standard reading of the responsibility-avoidance literature. The classic finding—that delegation reduces punishment of the principal and that this prospect motivates delegation—was established in dictator-game settings where the underlying choice is between a known fair and a known unfair allocation, the intermediary is human, and the principal can be held accountable specifically for choosing a particular intermediary (**???**). When the intermediary is an algorithm calibrated to mimic the principal and the underlying decision involves genuine uncertainty, none of the standard mechanisms through which delegation can shift responsibility—signal value of the delegation choice, the human-human punishment channel, the certainty of the unfair outcome—are available. What remains is a baseline outcome-based punishment, supplemented by an asymmetric desire to be the proximate cause of good outcomes when the moral stakes are salient.

**Implications, with explicit bounds.** Our findings speak to the design of accountability regimes governing algorithmic decision-making. The European Union's AI Act (**?**) and analogous regulatory frameworks elsewhere place increasing emphasis on assigning responsibility to human users of high-risk algorithmic systems. The implicit theory of behavior underlying these provisions is that holding humans accountable for algorithm-mediated decisions *disciplines* their use of those algorithms. Our results offer a behavioral complement to this view: when punishment is on the table, decision-makers in our setting do not simply delegate-and-blame the algorithm; they choose to take direct responsibility for the decision more often. From the perspective of efficiency, this need not be welfare-increasing: in domains where algorithms genuinely outperform humans, holding humans liable for outcomes may push decision-makers away from the more accurate technology. The trade-off between encouraging algorithmic adoption and assigning meaningful responsibility is, on this reading, sharper than the existing literature has suggested.

We note explicitly the bounds of this reading. Our evidence is a single online experiment with 322 UK Prolific participants on a stylised weight-prediction task, with stakes of £5 for Player B and a punishment range of £0–£2. The high-stakes algorithmic domains that motivate the regulatory debate—medical diagnosis, judicial risk assessment, hiring, credit allocation—differ from our setting in stake magnitude, professional context, expert training, the institutional structure of accountability, and the kind of population at the decision margin. We therefore view the result as identifying a behavioral channel that exists in principle and is detectable in our setting, not as a calibration of how big a lever accountability is in any specific applied domain. Mapping the magnitude of the effect against external benchmarks—e.g., physician adoption of diagnostic algorithms under malpractice exposure, or loan-officer deviation from credit-scoring algorithms under fair-lending audit—would require either domain-specific replication or instruments to which we do not have access. We flag this gap explicitly so that readers can calibrate the policy implication appropriately.

**Limitations.** Several caveats are in order. The belief measure we elicit is unincentivised and noisy; it conveys ordinal patterns reliably but cannot support fine-grained quantitative claims about Player A's anticipations. Two of the four belief scenarios are hypothetical conditional on the delegation decision, and all four are hypothetical in the *No-Punishment* condition; this further limits the inference, and we plan to report the H2 patterns separately for the payoff-relevant and the hypothetical cells once the analysis pipeline is in place. We use the strategy method for Player B's punishment decision, which is standard in this literature but may yield different absolute levels of punishment than a direct elicitation. Player A also makes the final prediction whether or not she ultimately decides to delegate, which equates the time cost of the two options but may signal her commitment differently than a design in which the prediction is skipped under delegation. We did not include a separate manipulation-comprehension check verifying that Player A internalised the punishment-on/off distinction; we relied on the prominent screen-level framing and on attention-check pass rates as indirect evidence, but a future replication should add this check. The absolute stakes are modest (£5 high payoff, £0–2 punishment range): we cannot speak to how the effect scales with stakes, and a stake-magnitude robustness check is the first item in the future-research list below. Our sample is drawn from UK Prolific participants, which limits external validity along the dimensions in which Prolific samples differ from broader populations and from professional decision-makers in the high-stakes algorithmic domains we discuss above.

A natural concern is whether the punishment possibility cues Player A to behave more conservatively for reasons unrelated to the substantive question—an experimenter-demand effect. We cannot rule this concern out from the data. The direction of any such effect is also ambiguous: a decision-maker who is reminded of being judged could plausibly delegate *more* (to shed responsibility) or *less* (to demonstrate effort and commitment); the literature does not unambiguously favor one over the other. We note two patterns weakly inconsistent with a strong demand reading—behavior does not track elicited beliefs as tightly as a pure-demand story would predict, and the sharp performance heterogeneity in the *Punishment* condition (better performers delegating less) is harder to motivate from demand alone—but we treat these as suggestive rather than dispositive. A direct test in the spirit of **?** would be the natural next step.

**Directions for future research.** Several extensions would strengthen the inference. A direct-method companion would test the robustness of Result 2 against the strategy-method elicitation, and a meaningfully larger sample would let us put a tight confidence interval on the small but persistent point estimate of insulation under delegation. A condition in which Player A is told explicitly that Player B is unaware of the equal-performance calibration would test whether the relative-performance signal is the binding margin distinguishing our results from **?**. Conditions that experimentally vary the algorithm's performance relative to Player A (above, equal, below) would map the generality of the process-ownership reading. A demand-effect probe along the lines of **?** would put a credible upper bound on the contribution of M4. A stake-magnitude robustness check, varying the £5/£1 high/low payoffs and the £0–£2 punishment range, would speak to whether the effect scales with stakes in the direction that policy-relevant settings would imply. Finally, replication in a different country, in a different decision domain (e.g., a medical-triage or hiring vignette), or with a sample of professional decision-makers rather than online crowd-workers would speak directly to the external-validity bounds we describe above.

Taken together, our results suggest a more textured story than the one that has emerged from the existing literature on responsibility avoidance and algorithm aversion. Within the bounds of the setting we study, the capacity to use algorithms as a tool for shifting responsibility appears limited, and the prospect of being held responsible—rather than encouraging delegation as a hedge—appears to push decision-makers towards direct engagement with the consequential choice. Whether the same channel operates in the high-stakes algorithmic domains that animate the regulatory debate is an empirical question that this study can frame but cannot, on its own, answer.

# Appendix

# A Additional Tables

## A.1 Condition balance — Player A

Table 2: Condition balance — Player A

| Variable | *Punishment* | *No-Punishment* | $p$-value |
|---|---|---|---|
| Age | 39.026 | 37.877 | 0.552 |
| Female | 0.487 | 0.494 | 1.000 |
| Socio-economic status | 5.179 | 5.321 | 0.564 |
| Went to uni | 0.575 | 0.556 | 0.928 |
| Technology score | 2.087 | 2.284 | 0.322 |
| Leadership position | 0.362 | 0.333 | 0.824 |

*Note:* Mean values by condition for Player A, with $p$-values from two-sided $t$-tests for continuous variables and Chi-squared tests for binary variables. *Socio-economic status* is self-reported on a 1–10 scale; *Went to uni* indicates a university degree; *Technology score* is constructed from weekly device usage, programming skills, cryptocurrency knowledge, technology use at work, and similar items; *Leadership position* indicates self-reported management or supervisory experience.

We do not separately report a balance table for Player B. Player B's punishment behavior is identified within-subject across the four scenarios of the strategy method, so balance across conditions is not strictly required for the punishment analysis.

## A.2 Full regression results — delegation decision

## A.3 Robustness — delegation decision

## A.4 Determinants of Player B's chosen punishment

## A.5 Robustness — Player B's chosen punishment

# B Bonus computation

Player A spent on average 12 minutes on the experiment and earned an average bonus of £2.49 on top of the Prolific show-up fee; Player B spent on average 8 minutes and earned an average bonus of £2.30. Bonuses are computed deterministically from the cleaned data per the experimental rules: random-round task payoff (expectation over the uniform draw across rounds 1–10); belief-about-own-score payoff (£0.30 times the probability mass placed on the realized score); belief-about-distribution payoff (expectation over the uniform random draw of which score is checked, paying £0.30 if the estimate is within 5 percentage points of the true population share); and—for Player A in the *Punishment* condition—expected punishment

Table 3: Determinants of the delegation decision — full controls

| | Dependent variable: *Delegation* | | |
|---|---|---|---|
| | *Full sample* | | *Punishment only* |
| | (1) | (2) | (3) |
| No-Punishment indicator | 0.677 | 0.757 | |
| | (0.320) | (0.334) | |
| Punishment beliefs: bad outcome | | | 0.937 |
| | | | (0.751) |
| Punishment beliefs: good outcome | | | 1.579 |
| | | | (0.741) |
| Task performance | | −0.192 | −0.609 |
| | | (0.128) | (0.236) |
| Age | | −0.004 | 0.064 |
| | | (0.015) | (0.030) |
| Female | | 0.350 | 1.334 |
| | | (0.369) | (0.793) |
| Socio-economic status | | −0.058 | −0.003 |
| | | (0.116) | (0.219) |
| Went to uni | | 0.091 | −0.433 |
| | | (0.364) | (0.707) |
| Technology score | | −0.078 | 0.059 |
| | | (0.153) | (0.283) |
| Leadership position | | 0.676 | −0.387 |
| | | (0.367) | (0.729) |
| Constant | −0.302 | 0.402 | −1.842 |
| | (0.226) | (0.930) | (2.103) |
| N | 161 | 159 | 60 |
| Pseudo $R^2$ | 0.020 | 0.049 | 0.246 |

*Note:* Logit estimates with robust (HC1) standard errors in parentheses. Two subjects in Column (2) are dropped due to missing socio-demographic data. Column (3) restricts to the *Punishment* condition and excludes subjects who failed the attention check on the belief-elicitation screen, as preregistered. The belief variables in Column (3) are within-subject differences in expected punishment between non-delegated and delegated decisions for each outcome. Following AEA editorial policy, significance stars are not reported in the table.

Table 4: Determinants of the delegation decision — robustness

| | Dependent variable: *Delegation* | | |
|---|---|---|---|
| | *Full sample* | | *Punishment* (no excl.) |
| | (1) | (2) | (3) |
| No-Punishment indicator | 0.771 | 0.677 | |
| | (0.336) | (0.320) | |
| Task performance | −0.277 | | −0.395 |
| | (0.149) | | (0.185) |
| Overconfidence | −0.103 | | |
| | (0.094) | | |
| Punishment beliefs: bad outcome (no del. − del.) | | | −0.277 |
| | | | (0.499) |
| Punishment beliefs: good outcome (no del. − del.) | | | 0.812 |
| | | | (0.529) |
| Controls | yes | no | yes |
| Attention-check excl. | yes (Player A) | n/a | no |
| Sample | Full | Full | Punishment |
| N | 159 | 161 | 78 |
| Pseudo $R^2$ | 0.055 | 0.020 | 0.114 |

*Note:* Robustness specifications for the delegation logit. Column (1) adds *overconfidence* (defined as the weighted-average belief about own performance minus actual performance) to the main full-sample specification. The treatment effect remains: *No-Punishment indicator* $p = 0.022$. *Overconfidence* is not significantly associated with delegation $p = 0.272$. Column (2) reproduces the unconditional treatment effect for completeness ($p = 0.034$; identical to the main Col (1)). Column (3) restricts to the *Punishment* condition and includes the within-subject belief differences but *does not* exclude subjects who failed the belief-screen attention check; the heterogeneity result is preserved: *Task performance* $p = 0.033$; *Punishment beliefs: good outcome* $p = 0.125$. Robust HC1 standard errors in parentheses. Following AEA editorial policy, significance stars are not reported in the table.

Table 5: Determinants of Player B's chosen punishment

| | Dependent variable: *Punishment* |
|---|---|
| | *Punishment* (£) |
| | (1) |
| Delegated | 0.001 |
| | (0.054) |
| Bad outcome | 0.260 |
| | (0.079) |
| Delegated × Bad outcome | −0.078 |
| | (0.052) |
| Age | −0.001 |
| | (0.006) |
| Female | −0.065 |
| | (0.143) |
| Socio-economic status | 0.072 |
| | (0.049) |
| Went to uni | −0.099 |
| | (0.136) |
| Technology score | 0.030 |
| | (0.058) |
| Leadership position | 0.060 |
| | (0.163) |
| Constant | 0.027 |
| | (0.344) |
| N (subject-by-cell) | 268 |
| Adj. $R^2$ | 0.034 |

*Note:* OLS estimates of Player B's chosen punishment on a delegation indicator, a bad-outcome indicator, their interaction, and socio-demographic controls. Standard errors clustered at the Player B level (in parentheses). Sample: Player B subjects in the *Punishment* condition who passed the punishment-elicitation attention check, expanded over the four (delegation, outcome) cells of the strategy method. Two-sided $p$-values: *Delegated* $p = 0.989$; *Bad outcome* $p = 0.001$; *Delegated* × *Bad outcome* $p = 0.137$. Following AEA editorial policy, significance stars are not reported.

Table 6: Determinants of Player B's chosen punishment — robustness

| | Dependent variable: *Punishment* | |
|---|---|---|
| | *Excl.* | *Incl. att. failers* |
| | (1) | (2) |
| Delegated | 0.001 | 0.032 |
| | (0.054) | (0.052) |
| Bad outcome | 0.260 | 0.256 |
| | (0.080) | (0.071) |
| Delegated × Bad outcome | −0.078 | −0.103 |
| | (0.052) | (0.058) |
| Player B belief about Player A perf. (wa) | 0.001 | −0.020 |
| | (0.037) | (0.032) |
| Age | −0.001 | 0.002 |
| | (0.006) | (0.005) |
| Female | −0.064 | −0.100 |
| | (0.140) | (0.128) |
| Socio-economic status | 0.072 | 0.083 |
| | (0.049) | (0.047) |
| Went to uni | −0.098 | −0.196 |
| | (0.142) | (0.140) |
| Technology score | 0.030 | 0.073 |
| | (0.059) | (0.053) |
| Leadership position | 0.061 | 0.083 |
| | (0.164) | (0.142) |
| Constant | 0.020 | −0.077 |
| | (0.374) | (0.328) |
| Attention-check excl. | yes | no |
| N (subject-by-cell) | 268 | 320 |
| Adj. $R^2$ | 0.030 | 0.077 |

*Note:* Punishment regression with Player B's belief about Player A's performance (*wa_difficulty*) added as a regressor. Column (1) excludes Player Bs who failed the punishment-screen attention check (preregistered); Column (2) keeps them. Standard errors clustered at the Player B level. The delegation, outcome, and interaction coefficients are stable across both specifications, and the belief regressor is not significantly associated with chosen punishment. H2 paired-test on the bad-outcome cell: with exclusion paired-$t$ $p = 0.153$, Wilcoxon $p = 0.533$ ($n = 67$); without exclusion paired-$t$ $p = 0.131$, Wilcoxon $p = 0.544$ ($n = 80$). Following AEA editorial policy, significance stars are not reported.

from the matched Player B's strategy-method choice in the realized cell, with £0.10 per cell of strategy-method punishment cost for Player B. The reported bonuses exclude the risk-elicitation MPL payoff (the raw-data column was not retained in the cleaned files); including it would add roughly £0.30–£0.50 per subject.

# C Ex-ante power calculations

The cell size of 80 subjects per role per condition was set to detect, at conventional power, the effect sizes anticipated from the responsibility-avoidance literature. We report the minimum detectable effect (MDE) for each preregistered test below, computed at $\alpha = 0.05$ (two-sided) and power = 0.80. For H1, the relevant test is a two-sample $z$-test of proportions (delegation rate in *Punishment* vs *No-Punishment*); the MDE is approximately invariant to the assumed baseline rate. For H2, the relevant test is a paired-mean-difference test on Player B's punishment in the bad-outcome cell; because the within-subject standard deviation $\sigma_d$ of the punishment difference is unknown ex ante, we report the MDE for a range of values of $\sigma_d$ expressed as multiples of the equivalence bound used in Result 2 of the body of the paper ($\delta_a =$ £0.20, equal to 10% of the maximum punishment of £2).

Table 7: Minimum detectable effects under the preregistered design

| Hypothesis | Assumption | Minimum detectable effect |
|---|---|---|
| H1 (delegation, two-proportion test) | Baseline $p_0 = 0.30$ | $\Delta = 0.216$ (21.6 pp) |
| H1 (delegation, two-proportion test) | Baseline $p_0 = 0.50$ | $\Delta = 0.215$ (21.5 pp) |
| H1 (delegation, two-proportion test) | Baseline $p_0 = 0.70$ | $\Delta = 0.179$ (17.9 pp) |
| H2 (paired, within-subject) | $\sigma_d = 0.5\,\delta_a =$ £0.10 | $\Delta =$ £0.031 |
| H2 (paired, within-subject) | $\sigma_d = 1\,\delta_a =$ £0.20 | $\Delta =$ £0.063 |
| H2 (paired, within-subject) | $\sigma_d = 2\,\delta_a =$ £0.40 | $\Delta =$ £0.125 |
| H2 (paired, within-subject) | $\sigma_d = 3\,\delta_a =$ £0.60 | $\Delta =$ £0.188 |
| H2 (paired, within-subject) | $\sigma_d = 4\,\delta_a =$ £0.80 | $\Delta =$ £0.251 |

*Note:* Both tests assume $n = 80$ per role per condition, $\alpha = 0.05$ two-sided, power = 0.80. For H2, $\sigma_d$ is the within-subject standard deviation of the punishment difference (delegated minus non-delegated, conditional on bad outcome) and is expressed as a multiple of the anchor effect $\delta_a =$ £0.20 (the equivalence bound used in Result 2 in the body of the paper, equal to 10% of the maximum punishment of £2). $\Delta$ is the corresponding minimum detectable effect in pounds.

# D Preregistration and deviations

The experiment was preregistered at AsPredicted #133980, available at aspredicted.org/hs9az8.pdf and registered on 30 May 2023 prior to the main wave of data collection. The preregistration commits to the two research questions stated in the body as H1 and H2, the sample-size target (320 subjects total, 80 per role per condition), the desktop-only restriction, the use of the strategy method to elicit Player B's punishment, and a set of primary statistical tests (parametric

$t$-tests and non-parametric MWU and Kolmogorov–Smirnov tests for both comparisons; probit regressions for the delegation decision and linear regressions for punishment behavior with treatment, beliefs, and socio-demographics as predictors).

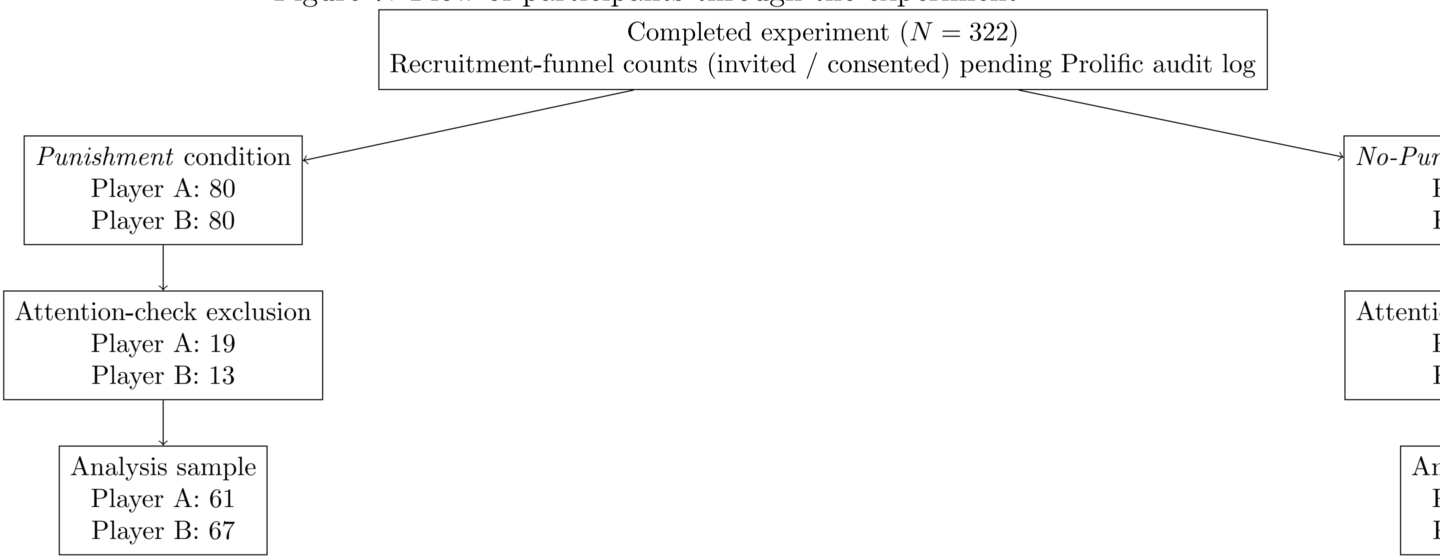
The preregistration also notes that a pilot of 25 observations had been collected in the *Punishment* treatment prior to registration; these observations are pooled with the main wave in all analyses, as preregistered. Two secondary analyses are pre-stated in the registry: the role of own-performance and beliefs about own performance (developed as M2 in Section **??**), and the consistency of Player A's beliefs about expected punishment with her delegation decision (developed as M3 in Section **??**).

The table below summarizes every analysis reported in the body of the paper, the corresponding preregistered specification, and any deviation.

# E Flow of participants

The flow of participants through the experiment, by condition and role, is shown in Figure **??**. Recruitment-funnel counts (invited / consented) are pending the Prolific audit log.

Figure 7: Flow of participants through the experiment




# F Additional Figures

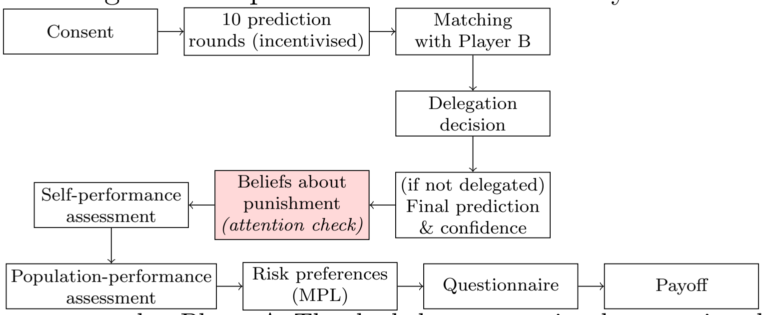
## F.1 Experimental timelines

## F.2 Punishment frequencies

Table 8: Preregistered tests, reported tests, and deviations

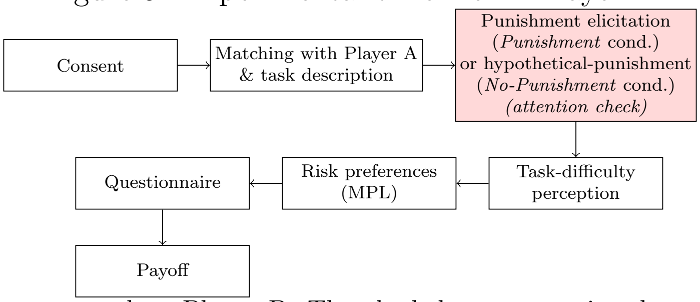
| Preregistered specification | As reported | Deviation / rationale |
|---|---|---|
| H1: between-treatment test of delegation rates (*Punishment* vs *No-Punishment*); preregistered tests are parametric $t$-test, MWU, KS. Direction not specified in the registry. | Pearson Chi-squared and Fisher's exact, two-sided ($p = 0.033$ / $p = 0.041$). | Test family substituted (Chi-squared and Fisher's exact are the standard tests for a binary outcome; the preregistered $t$-test, MWU and KS tests on a binary outcome are equivalent to the proportion test). The realized direction is opposite to that anticipated by the responsibility-avoidance literature on which we drew. |
| H2: paired test of Player B punishment in the bad-outcome cell, delegated vs non-delegated, attention-check passers only | Paired $t$-test ($p = 0.153$); Wilcoxon signed-rank ($p = 0.533$). | As preregistered. |
| Probit / logit of delegation on condition + controls (Table **??**, Cols (1)–(2)) | Logit, as reported. | As preregistered. (Probit and logit are equivalent for our purposes; we report logit for ease of marginal-effect interpretation.) |
| Logit on *Punishment* subsample with belief differences (Col (3)) | As reported. | Specification preregistered; sample restriction follows the preregistered attention-check rule. |
| Pre-stated secondary analysis: own-performance and beliefs about own performance | Reported as M2 in Section **??**. | As preregistered (secondary). |
| Pre-stated secondary analysis: beliefs about expected punishment vs delegation decision | Reported as M3 in Section **??**. | As preregistered (secondary). |
| Mechanism analyses M1 (process ownership) and M4 (experimenter demand) (Section **??**) | As reported. | *Not preregistered.* Treated as exploratory. |

Figure 8: Experimental timeline — Player A



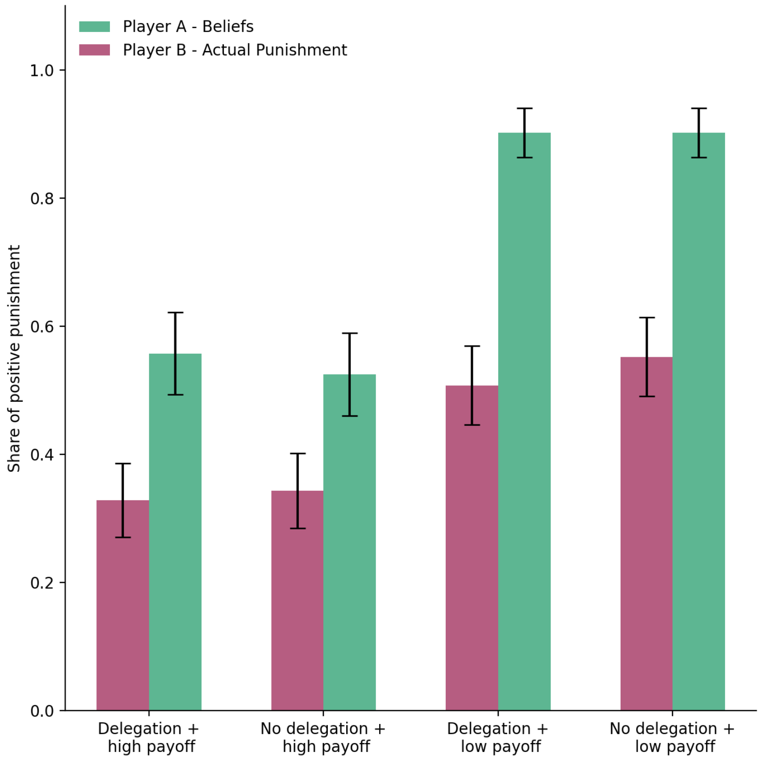

*Note:* Screen sequence presented to Player A. The shaded screen carries the attention check used to identify subjects excluded from belief-related analyses. The final-prediction-and-confidence screen is skipped if Player A delegates to the algorithm.

Figure 9: Experimental timeline — Player B




*Note:* Screen sequence presented to Player B. The shaded screen carries the attention check used to identify subjects excluded from punishment-related analyses. In the *No-Punishment* condition, no actual punishment is implemented; the elicitation is hypothetical and serves to maintain comparability of the screen sequence across conditions.

Figure 10: Frequencies of punishment and beliefs about punishment




*Note:* Share of Player Bs choosing strictly positive punishment in each of the four scenarios (solid bars) and share of Player As assigning a strictly positive expected punishment in each scenario (hollow bars).